AP CALCULUS BC · EXAMINATION REPOSITORY

Unit 6

Integration & Accumulation of Change

100-Question Priority-Starred Master Set

- Questions 1–90: Multiple Choice (Full Priority Ratings · 4 Questions Per Page)
- Questions 91–100: Free Response with Official AP Scoring Rubrics (9 Pts Each)
- Integrated AP Semicircle/Piecewise Linear Vector Graphs & Riemann Sum Data Tables
- Complete Answer Key and Detailed Worked Solutions at the End

OFFICIAL AP CED ALIGNED · MEDIUM LEVEL · RIGOROUSLY VERIFIED

Examination Structure

This paper contains 100 continuously numbered questions designed to reflect the rigorous analytical and conceptual standard of the AP Calculus BC examination, integrating authentic tables and vector area accumulation graphs. All answers, worked solutions, and scoring rubrics appear at the end of the document.

⭐ Priority Rating System

★★★★★ VERY IMPORTANT	High-frequency exam concepts: FTC (Parts 1 & 2), u-substitution, Integration by Parts, Improper Integrals, Partial Fractions, accumulation area graphs.
★★★ MANDATORY	Essential skills: standard antiderivatives, definite integral properties, Riemann sum table approximations, algebraic rewriting.
★★ IMPORTANT	Foundational practice: basic power and exponential rules, elementary summation forms.

- **Questions 1–90:** Multiple choice, four options, exactly one correct answer (4 questions per page).
- **Questions 91–100:** Free response, 9 points each, with analytical setups and rubrics.
- **Visual Features:** Vector SVG geometric area graphs, velocity/flow data tables, and structured solution steps.
- **Calculator Policy:** No calculator required (Active analytical calculus).
- **Answer Section:** Begins after Question 100 — answer key, then full MCQ solutions, then FRQ solutions with rubrics.

Topic Coverage (Unit 6.1 – 6.13)

6.1 Accumulations of Change	6.2 Riemann Sums Approximations
6.3 Riemann Sums in Summation Notation	6.4 FTC and Accumulation Functions
6.5 Accumulation Functions Involving Area	6.6 Properties of Definite Integrals
6.7 FTC Part 2 (Definite Integrals)	6.8 Antiderivatives & Indefinite Integrals
6.9 Integrating Using Substitution	6.10 Long Division & Completing Square
6.11 BC: Integration by Parts	6.12 BC: Linear Partial Fractions
6.13 BC: Evaluating Improper Integrals	Inverse Trigonometric Forms

Study Tip: The Fundamental Theorem of Calculus is the golden thread connecting derivatives and integrals. When facing accumulation functions $g(x) = \int^{x} f(t)\,dt$, always remember that $g'(x) = f(x)$ and $g''(x) = f'(x)$.

ITEM 01

EASY · SINGLE BEST CHOICE ★★ IMPORTANT

A water tank leaks at a rate of $r(t) = 3\sqrt{t}$ gallons per hour, where t is measured in hours. How many gallons of water leak from the tank between $t = 1$ and $t = 4$ hours?

A 7
B 21
C 14
D 18

ITEM 02

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

Let $f(x) = x^2 + 1$. What is the approximation of $\int_0^6 f(x)\,dx$ using a Left Riemann sum with 3 equal subintervals?

x	0	2	4	6
f(x)	1	5	17	37

A 23
B 70
C 118
D 46

ITEM 03

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The table below gives values of a continuous function f at selected values of x.

x: 1, 3, 4, 7
f(x): 4, 6, 5, 2

What is the approximation of $\int_1^7 f(x)\,dx$ using a Right Riemann sum with the three subintervals indicated by the table?

t (seconds)	0	2	5	8	10
v(t) (ft/sec)	12	18	25	20	14

A 29
B 17
C 23
D 34

ITEM 04

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

What is the approximation of $\int_0^4 (2x + 3)\,dx$ using a Midpoint Riemann sum with 2 subintervals of equal length?

A 14
B 32
C 28
D 26

ITEM 05

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The rate at which oil flows through a pipeline is recorded in the table below at selected times t (hours):

t: 0, 2, 5, 8
r(t) (barrels/hr): 10, 16, 22, 30

Using a trapezoidal sum with the three subintervals indicated by the table, what is the estimated total number of barrels that flow through the pipeline from t = 0 to t = 8?

t (hours)	0	3	7	10	12
R(t) (gal/hr)	8.0	7.2	5.6	4.0	3.2

A 161
B 156
C 178
D 142

ITEM 06

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let f be a function that is continuous and strictly increasing on the closed interval [a, b]. Which of the following Riemann sum approximations of $\int_a^b f(x)\,dx$ is guaranteed to strictly underestimate the exact value of the integral?

A Right Riemann sum
B Midpoint Riemann sum
C Trapezoidal sum
D Left Riemann sum

ITEM 07

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A function f is twice-differentiable on [1, 5] with $f''(x) > 0$ for all x in [1, 5]. If T is the trapezoidal sum approximation of $\int_1^5 f(x)\,dx$ using 4 subintervals of equal length, which of the following statements must be true?

A $T > \int_1^5 f(x)\,dx$
B $T < \int_1^5 f(x)\,dx$
C $T = \int_1^5 f(x)\,dx$
D T cannot be compared to the integral without knowing whether $f'(x) > 0$

ITEM 08

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

A tank initially holds 50 gallons of liquid at time t = 0. Liquid is pumped into the tank at a rate of E(t) = 4t gallons per minute and drains out at a rate of L(t) = 2t gallons per minute for $0 \le t \le 5$. What is the total volume of liquid in the tank at t = 4 minutes?

A 16 gallons
B 82 gallons
C 34 gallons
D 66 gallons

ITEM 09

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

Let $f(x) = 1/x$ on the interval $[1, 7]$. What is the Midpoint Riemann sum approximation of $\int_1^7 (1/x)\,dx$ using 3 subintervals of equal length?

A 11/6
B 11/12
C 7/6
D 2

ITEM 10

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the approximation of $\int_0^{\pi} \sin x\,dx$ using the Trapezoidal Rule with 4 subintervals of equal length?

A $\pi(\sqrt{2}+2)/4$
B $\pi\sqrt{2}/4$
C $\pi(2\sqrt{2}+1)/8$
D $\pi(\sqrt{2}+1)/4$

ITEM 11

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following definite integrals is equivalent to the limit of the Riemann sum $\lim_{n\to\infty} \sum_{k=1}^{n} (1+3k/n)^2 \cdot (3/n)$?

A $\int_1^4 x^2\,dx$
B $\int_0^3 x^2\,dx$
C $\int_1^4 (1+3x)^2\,dx$
D $3\int_1^4 x^2\,dx$

ITEM 12

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\lim_{n\to\infty} \sum_{k=1}^{n} \cos(\pi k/(2n)) \cdot (\pi/(2n))$?

A 0
B $\pi/2$
C 1
D 2

ITEM 13

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

What is the value of $\lim_{n\to\infty} (1/n) \sum_{k=1}^{n} \sqrt{(k/n)}$?

A 1/2
B 2/3
C 1/3
D 3/2

ITEM 14

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following definite integrals is represented by the limit $\lim_{n\to\infty} \sum_{k=1}^{n} [\,2 + 5k/n\,]^3 \cdot (5/n)$?

A $\int_0^5 (2+x)^3\,dx$
B $\int_2^7 x^3\,dx$
C $5\int_2^7 x^3\,dx$
D $\int_0^5 x^3\,dx$

ITEM 15

EASY · SINGLE BEST CHOICE ★★★ MANDATORY

For a continuous function f on [0, 4], a Left Riemann sum with 4 equal subintervals gives an approximation of L = 18, and a Right Riemann sum with the same 4 equal subintervals gives R = 26. What is the trapezoidal sum approximation T of $\int_0^4 f(x)\,dx$ using these 4 subintervals?

A 22
B 20
C 24
D 44

ITEM 16

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following definite integrals is equivalent to $\lim_{n\to\infty} \sum_{k=1}^{n} e^{2k/n} \cdot (2/n)$?

A $\int_0^1 e^{2x}\,dx$
B $2\int_0^2 e^{x}\,dx$
C $\int_0^2 e^{x}\,dx$
D $\int_1^2 e^{2x}\,dx$

ITEM 17

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

If f is a continuous function such that $f''(x) < 0$ for all x in [a, b], which of the following inequalities correctly compares the Midpoint Riemann sum M, the Trapezoidal sum T, and the exact definite integral $I = \int_a^b f(x)\,dx$ (all using the same n subintervals)?

A $T < I < M$

B $M < I < T$

C $I < T < M$

D $M < T < I$

ITEM 18

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\lim_{n\to\infty} \sum_{k=1}^{n} [\,1 / (1 + (k/n)^2)\,] \cdot (1/n)$?

A $\pi/4$

B $\pi/2$

C $\ln 2$

D 1

ITEM 19

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

The velocity v(t) of a particle moving along a line is given in meters per second in the table below:

t (seconds): 0, 3, 6, 9
v(t) (m/s): 2, 5, 7, 8

Using the Trapezoidal Rule with the 3 subintervals shown, what is the estimated distance (in meters) traveled by the particle from t = 0 to t = 9?

A 51 meters

B 68 meters

C 34 meters

D 45 meters

ITEM 20

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following limits of a Riemann sum is equal to the definite integral $\int_3^5 \ln(x)\,dx$?

A $\lim_{n\to\infty} \sum_{k=1}^{n} \ln(3 + k/n) \cdot (2/n)$

B $\lim_{n\to\infty} \sum_{k=1}^{n} \ln(3 + 2k/n) \cdot (2/n)$

C $\lim_{n\to\infty} \sum_{k=1}^{n} \ln(2k/n) \cdot (2/n)$

D $\lim_{n\to\infty} \sum_{k=1}^{n} \ln(3 + 2k/n) \cdot (1/n)$

ITEM 21

EASY · SINGLE BEST CHOICE ★★★ MANDATORY

If $g(x) = \int_{2}^{x} \sqrt{(t^3 + 1)}\, dt$, then $g'(2) =$

A 0
B 9
C 3
D 12

ITEM 22

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

$d/dx \left[\int_{0}^{x^2} \cos(t)\, dt \right] =$

A $\cos(x^2)$
B $2x \cos(x^2)$
C $-2x \sin(x^2)$
D $2x \sin(x^2)$

ITEM 23

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

$d/dx \left[\int_{5}^{x} e^{t^2}\, dt \right] =$

A $-e^{x^2}$
B e^{x^2}
C $2x\, e^{x^2}$
D $-2x\, e^{x^2}$

ITEM 24

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $h(x) = \int_{3}^{3x} \ln(t)\, dt$ for $x > 0$. What is $h'(e/3)$?

A 1
B e
C 3e
D 3

ITEM 25

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $x > 0$, $\frac{d}{dx}\left[\int_{x^3}^{2x} \frac{1}{t+1}\,dt\right] =$

A $\frac{3x^2}{x^3+1} - \frac{2}{2x+1}$

B $\frac{1}{x^3+1} - \frac{1}{2x+1}$

C $\frac{3x^2}{x^3+1} + \frac{2}{2x+1}$

D $\frac{3x^2-2}{x^3-2x+1}$

ITEM 26

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $g(x) = \int_{3x}^{x} (t^2 - 2t)\,dt$. What is an equation of the line tangent to the graph of g at $x = 3$?

A $y = 3x$

B $y = 3x + 9$

C $y = 3x - 9$

D $y = (x^2 - 2x)(x - 3)$

ITEM 27

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

If $f(x) = \int_{0}^{x} t\sin(t)\,dt$, then $f''(\pi/2) =$

A 0

B $\pi/2$

C -1

D 1

ITEM 28

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $x > 0$, let $F(x) = \int_{1}^{x} (t^2 - 4)\,dt$. At what value of x does F achieve a relative minimum?

A $x = 1$

B $x = 2$

C $x = 4$

D $x = -2$

ITEM 29

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

$d/dx\,[\,\int_{0}^{\sin x} e^{t^2}\,dt\,] =$

A $e^{\sin^2 x}$

B $\cos x \cdot e^{\sin^2 x}$

C $-\cos x \cdot e^{\sin^2 x}$

D $2\sin x \cos x \cdot e^{\sin^2 x}$

ITEM 30

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\lim_{x \to 0}\,[\,(\int_{0}^{x} \cos(t^2)\,dt)\,/\,x\,]$?

A 1

B 0

C π

D undefined

ITEM 31

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The continuous function f is defined on the closed interval [−2, 6]. The graph of f consists of four line segments: from (−2, 0) to (0, 4), from (0, 4) to (2, 0), from (2, 0) to (4, −4), and from (4, −4) to (6, 0). Let $g(x) = \int_{0}^{x} f(t)\,dt$. What is the value of g(4)?

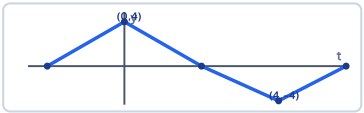


FIGURE 6.4: GRAPH OF Y = F(T) ON [−2, 6] FOR QUESTIONS 31–35

A 8

B 0

C 4

D −4

ITEM 32

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Using the same function f from the previous question, where the graph on [−2, 0] is a line segment from (−2, 0) to (0, 4) and $g(x) = \int_{0}^{x} f(t)\,dt$, what is the value of g(−2)?

A 4

B 0

C −4

D −8

ITEM 33

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For the function f whose graph is above the x-axis on (−2, 2) and below the x-axis on (2, 6), let $g(x) = \int_0^x f(t)\,dt$ on [−2, 6]. On which open interval is g strictly increasing?

A (−2, 2)
B (0, 4)
C (2, 6)
D (−2, 0)

ITEM 34

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

With f defined on [−2, 6] as above and $g(x) = \int_0^x f(t)\,dt$, at what value of x does g attain its absolute maximum on [−2, 6]?

A x = 0
B x = 2
C x = 4
D x = 6

ITEM 35

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For the function f defined by line segments from (−2, 0) to (0, 4) to (4, −4) to (6, 0), on which interval is the graph of $g(x) = \int_0^x f(t)\,dt$ concave down?

A (0, 4)
B (2, 6)
C (−2, 0)
D (4, 6)

ITEM 36

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following values of x corresponds to an x-coordinate of a point of inflection of the graph of $g(x) = \int_0^x f(t)\,dt$, where f increases on (−2, 0), decreases on (0, 4), and increases on (4, 6)?

A x = 2
B x = 0
C x = −2
D x = 1

ITEM 37

EASY · SINGLE BEST CHOICE ★★★ MANDATORY

Let $h(x) = 5 + \int_1^x f(t)\,dt$, where f is a continuous function. If $\int_1^4 f(t)\,dt = 9$, what is the value of h(4)?

A 9
B 4
C −4
D 14

ITEM 38

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The graph of a continuous function f on [0, 4] is a semicircle of radius 2 centered at (2, 0) situated above the x-axis. If $g(x) = \int_0^x f(t)\,dt$, what is the value of g(4)?

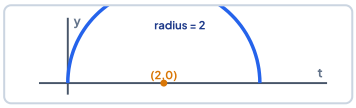


FIGURE 6.5: UPPER SEMICIRCLE OF RADIUS 2 ON [0, 4]

A 4π
B π
C 2π
D 8

ITEM 39

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Let $g(x) = \int_0^x f(t)\,dt$ for $x > 0$. If f(t) is positive and strictly decreasing for all $t > 0$, which of the following describes the graph of g for $x > 0$?

A Increasing and concave down
B Increasing and concave up
C Decreasing and concave down
D Decreasing and concave up

ITEM 40

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $H(x) = \int_0^x (e^t - 3)\,dt$. On what interval is H strictly decreasing?

A $(\ln 3, \infty)$
B $(0, 3)$
C $(-\infty, 3)$
D $(-\infty, \ln 3)$

ITEM 41

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

If $\int_{1}^{5} f(x)\,dx = 10$ and $\int_{3}^{5} f(x)\,dx = 4$, what is the value of $\int_{1}^{3} (2f(x) + 3)\,dx$?

A 15 **B** 12
C 18 **D** 21

ITEM 42

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Given that $\int_{0}^{4} f(x)\,dx = 7$ and $\int_{6}^{4} f(x)\,dx = 3$, what is the value of $\int_{0}^{6} f(x)\,dx$?

A 10 **B** 4
C −4 **D** 21

ITEM 43

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the value of $\int_{-2}^{2} (x^5 - 3x^3 + 4\sin x + 5)\,dx$?

A 0 **B** 10
C 40 **D** 20

ITEM 44

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

If f is an even continuous function such that $\int_{0}^{3} f(x)\,dx = 8$, what is the value of $\int_{-3}^{3} (f(x) - 2)\,dx$?

A 4 **B** 16
C 10 **D** 0

ITEM 45

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves along the x-axis with velocity $v(t) = 3t^2 - 2t$ for $t \geq 0$. If the position of the particle at $t = 1$ is $x(1) = 6$, what is its position at $t = 3$?

A 18 **B** 30

C 24 **D** 12

ITEM 46

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the value of $\int_0^4 |x - 2|\,dx$?

A 0 **B** 2

C 4 **D** 8

ITEM 47

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let f be defined piecewise by $f(x) = \{ 2x \text{ for } x < 1;\ 3x^2 \text{ for } x \geq 1 \}$. What is the value of $\int_0^2 f(x)\,dx$?

A 8 **B** 7

C 9 **D** 6

ITEM 48

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

What is the average value of $f(x) = 3x^2 - 2x$ on the closed interval $[0, 3]$?

A 18 **B** 6

C 9 **D** 3

ITEM 49

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\int_{0}^{\pi/4} (\sec^2 x + e^{2x})\,dx$?

A $(1 + e^{\pi/2}) / 2$

B $1 + e^{\pi/2}$

C $(e^{\pi/2} - 1) / 2$

D $1 + 2e^{\pi/2}$

ITEM 50

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the value of $\int_{1}^{e} ((2/x) + 3x^2)\,dx$?

A $e^3 + 1$

B $e^3 + 2$

C $e^3 - 1$

D $2e^3 + 1$

ITEM 51

EASY · SINGLE BEST CHOICE ★★★ MANDATORY

$\int ((x^3 - 4x^2 + 2) / x^2)\,dx =$

A $x^2/2 - 4x + 2/x + C$

B $(x^4/4 - 4x^3/3 + 2x) / (x^3/3) + C$

C $x^2/2 - 4x - 2/x + C$

D $x^2/2 - 4x - 2x + C$

ITEM 52

EASY · SINGLE BEST CHOICE ★★ IMPORTANT

$\int 5^x\,dx =$

A $5^x \ln 5 + C$

B $5^x / \ln 5 + C$

C $5^(x+1) / (x + 1) + C$

D $5^x + C$

ITEM 53

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

$\int (1 / \sqrt{(1 - 9x^2)})\, dx =$

A $\arcsin(3x) + C$

B $(1/9)\arcsin(3x) + C$

C $(1/3)\arctan(3x) + C$

D $(1/3)\arcsin(3x) + C$

ITEM 54

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

$\int (1 / (x^2 + 16))\, dx =$

A $(1/4)\arctan(x/4) + C$

B $\arctan(x/4) + C$

C $(1/16)\arctan(x/16) + C$

D $(1/4)\arcsin(x/4) + C$

ITEM 55

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

$\int \tan^2 x\, dx =$

A $(1/3)\tan^3 x + C$

B $\sec^2 x - x + C$

C $\tan x - x + C$

D $\tan x + x + C$

ITEM 56

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

If $dy/dx = 4x^3 + 2x - 1$ and $y(1) = 5$, what is the particular solution $y(x)$?

A $y = x^4 + x^2 - x + 5$

B $y = x^4 + x^2 - x + 4$

C $y = 12x^2 + 2$

D $y = x^4 + x^2 - x + 3$

ITEM 57

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

$\int ((2\sqrt{x} - 3) / \sqrt{x})\, dx =$

A $2x - 3\sqrt{x} + C$

B $2x - (3/2)\sqrt{x} + C$

C $x - 6\sqrt{x} + C$

D $2x - 6\sqrt{x} + C$

ITEM 58

EASY · SINGLE BEST CHOICE ★★ IMPORTANT

$\int \csc(4x)\cot(4x)\, dx =$

A $(1/4)\csc(4x) + C$

B $-(1/4)\csc(4x) + C$

C $-4\csc(4x) + C$

D $-(1/4)\sec(4x) + C$

ITEM 59

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

What is the value of $\int_1^2 (3 / x^4)\, dx$?

A 7/8

B −7/8

C 15/16

D 7/24

ITEM 60

MODERATE · SINGLE BEST CHOICE ★★★ MANDATORY

What is the value of $\int_0^{\pi/3} \sec x \tan x\, dx$?

A 2

B $\sqrt{3}$

C 1

D 0

ITEM 61

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

$\int x(x^2+4)^5\,dx =$

A $(1/12)(x^2+4)^6 + C$

B $(1/6)(x^2+4)^6 + C$

C $(1/3)(x^2+4)^6 + C$

D $(1/12)(x^2+4)^5 + C$

ITEM 62

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Using the substitution $u = 2x + 1$, the definite integral $\int_0^4 \sqrt{(2x+1)}\,dx$ is equivalent to which of the following?

A $\int_1^9 \sqrt{u}\,du$

B $(1/2)\int_0^4 \sqrt{u}\,du$

C $2\int_1^9 \sqrt{u}\,du$

D $(1/2)\int_1^9 \sqrt{u}\,du$

ITEM 63

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\int_0^1 x e^{x^2}\,dx$?

A $e - 1$

B $(e-1)/2$

C $(e+1)/2$

D $e/2$

ITEM 64

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

$\int \sin^4 x \cos x\,dx =$

A $(1/5)\cos^5 x + C$

B $-(1/5)\sin^5 x + C$

C $(1/5)\sin^5 x + C$

D $4\sin^3 x \cos^2 x + C$

ITEM 65

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $x > 0$, $\int ((\ln x)^3 / x)\,dx =$

A $(\ln x)^4 + C$

B $(1/4)(\ln x)^4 + C$

C $(1/3)(\ln x)^3 + C$

D $(3(\ln x)^2) / x^2 + C$

ITEM 66

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

$\int \tan^3 x \sec^2 x\,dx =$

A $(1/4)\sec^4 x + C$

B $(1/3)\tan^3 x + C$

C $(1/4)\tan^4 x + C$

D $\tan^4 x + C$

ITEM 67

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\int_0^{\sqrt{3}} (x / \sqrt{(x^2+1)})\,dx$?

A 1

B 2

C 1/2

D $\sqrt{3}$

ITEM 68

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $x > 1$, $\int x\sqrt{(x-1)}\,dx =$

A $(2/3)(x-1)^{(3/2)} + C$

B $(1/2)x^2 \cdot (2/3)(x-1)^{(3/2)} + C$

C $(2/5)(x-1)^{(5/2)} - (2/3)(x-1)^{(3/2)} + C$

D $(2/5)(x-1)^{(5/2)} + (2/3)(x-1)^{(3/2)} + C$

ITEM 69

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

If f is a differentiable function with f(1) = 2, f(4) = 8, and f(x) > 0 for all x, what is the value of $\int_1^4 (f'(x) / f(x))\,dx$?

A ln 6
B ln 4
C 4
D 6

ITEM 70

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A differentiable function g satisfies g(0) = 1 and g(2) = 3 with g' continuous. What is the value of $\int_0^2 g'(x)\,[g(x)]^2\,dx$?

A 8
B 26
C 26/3
D 13/3

ITEM 71

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $x \neq -1$, $\int ((x^2 + 1) / (x + 1))\,dx =$

A $x^2/2 + x + \ln|x + 1| + C$
B $x - \ln|x + 1| + C$
C $x^2/2 - x + 2\ln|x + 1| + C$
D $(x^3/3 + x) / (x^2/2 + x) + C$

ITEM 72

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

For $x \neq -2$, $\int ((3x + 5) / (x + 2))\,dx =$

A $3x + \ln|x + 2| + C$
B $3\ln|x + 2| + C$
C $3x - 5\ln|x + 2| + C$
D $3x - \ln|x + 2| + C$

ITEM 73

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

$\int (1 / (x^2 + 4x + 5))\, dx =$

A $\ln|x^2 + 4x + 5| + C$

B $\arctan(x + 2) + C$

C $(1/5)\arctan((x + 2)/5) + C$

D $\arctan(x + 4) + C$

ITEM 74

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

$\int (1 / (x^2 - 6x + 13))\, dx =$

A $(1/2)\arctan((x - 3)/2) + C$

B $\arctan((x - 3)/2) + C$

C $(1/4)\arctan((x - 3)/4) + C$

D $(1/2)\ln|(x - 3)^2 + 4| + C$

ITEM 75

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $-5 < x < 1$, $\int (1 / \sqrt{(5 - 4x - x^2)})\, dx =$

A $(1/3)\arcsin((x + 2)/3) + C$

B $\arctan((x + 2)/3) + C$

C $\ln|x + 2 + \sqrt{(5 - 4x - x^2)}| + C$

D $\arcsin((x + 2)/3) + C$

ITEM 76

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\int_0^1 ((2x^2 + 3x) / (x + 1))\, dx$?

A $2 + \ln 2$

B $2 - \ln 2$

C $1 - \ln 2$

D $2 - 2\ln 2$

ITEM 77

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

$\int ((2x + 5) / (x^2 + 1))\, dx =$

A $2\ln(x^2 + 1) + 5\arctan x + C$

B $\ln(x^2 + 1) + (5/2)\arctan x + C$

C $\ln(x^2 + 1) + 5\arctan x + C$

D $2x\ln(x^2 + 1) + 5\arctan x + C$

ITEM 78

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\int_{-11}^{1} (1 / (x^2 + 2x + 2))\, dx$?

A $\pi/4$

B $\arctan(1)$

C $\ln 5$

D $\arctan(2)$

ITEM 79

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

$\int ((x^3 + 2) / (x^2 + 1))\, dx =$

A $x^2/2 - (1/2)\ln(x^2 + 1) + 2\arctan x + C$

B $x^2/2 - \ln(x^2 + 1) + 2\arctan x + C$

C $x^2/2 + (1/2)\ln(x^2 + 1) + 2\arctan x + C$

D $x^2 - (1/2)\ln(x^2 + 1) + \arctan x + C$

ITEM 80

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\int_{1}^{2} (1 / \sqrt{(4x - x^2)})\, dx$?

A $\pi/3$

B $\pi/6$

C $\pi/4$

D $-\pi/6$

ITEM 81

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

$\int x\,e^{2x}\,dx =$

A $(1/2)\,x\,e^{2x} + (1/4)\,e^{2x} + C$

B $(1/2)\,x\,e^{2x} - (1/4)\,e^{2x} + C$

C $x\,e^{2x} - e^{2x} + C$

D $(1/2)\,x^2\,e^{2x} + C$

ITEM 82

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\int_1^e x \ln x\,dx$?

A $(e^2 - 1)/4$

B $e^2/2$

C $(e^2 + 3)/4$

D $(e^2 + 1)/4$

ITEM 83

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

$\int x^2 \sin x\,dx =$

A $-x^2\cos x + 2x\sin x + 2\cos x + C$

B $-x^2\cos x - 2x\sin x + 2\cos x + C$

C $x^2\cos x - 2x\sin x - 2\cos x + C$

D $-x^2\cos x + 2x\sin x - 2\cos x + C$

ITEM 84

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

$\int \arctan x\,dx =$

A $1/(1 + x^2) + C$

B $x\arctan x - \ln(1 + x^2) + C$

C $x\arctan x - (1/2)\ln(1 + x^2) + C$

D $x\arctan x + (1/2)\ln(1 + x^2) + C$

ITEM 85

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $x > 3$, $\int (1 / (x^2 - 5x + 6))\, dx =$

A $\ln|(x-3)/(x-2)| + C$

B $\ln|(x-2)/(x-3)| + C$

C $\ln|(x-2)(x-3)| + C$

D $(1/2)\ln|x^2 - 5x + 6| + C$

ITEM 86

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of $\int_4^5 (1 / (x^2 - 9))\, dx$?

A $(1/3)\ln(7/4)$

B $(1/6)\ln(7/4)$

C $(1/6)\ln(4/7)$

D $\ln(7/4)$

ITEM 87

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

For $x > -1$, $\int ((x+2) / (x+1)^2)\, dx =$

A $\ln|x+1| + 1/(x+1) + C$

B $\ln|x+1| - 2/(x+1) + C$

C $(1/2)\ln|(x+1)^2| + C$

D $\ln|x+1| - 1/(x+1) + C$

ITEM 88

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of the improper integral $\int_1^\infty (1 / x^3)\, dx$?

A 1

B 2

C 1/2

D divergent

ITEM 89

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of the improper integral $\int_0^\infty e^{-3x}\,dx$?

A 1/3
B 3
C 1
D divergent

ITEM 90

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the value of the improper integral $\int_0^4 (1/\sqrt{x})\,dx$?

A 2
B 4
C 8
D divergent

Question 91: Area & Accumulation Function from a Graph ★★★★★ VERY IMPORTANT 9 Points

The continuous function f is defined on the closed interval [−4, 6]. The graph of f consists of a semicircle of radius 2 centered at (0, 0) and three line segments: from (−4, 0) to (−2, 0), from (2, 0) to (4, 4), and from (4, 4) to (6, 0). The semicircle is situated above the x-axis on [−2, 2]. Let $g(x) = \int_{-2}^{x} f(t)\,dt$.

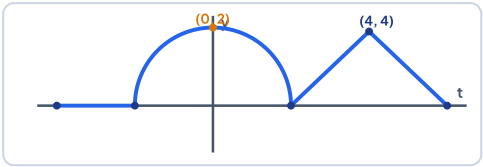


FIGURE FRQ-1: GRAPH OF CONTINUOUS FUNCTION F(T) ON [−4, 6] (SEMICIRCLE & LINE SEGMENTS)

(a) [2 Points] Find g(−4) and g(2). Show the geometric setups that lead to your answers.

(b) [2 Points] Find g'(4) and determine g"(4), or explain why it does not exist.

(c) [2 Points] Find all values of x in (−4, 6) at which g has a relative minimum. Justify your answer.

(d) [3 Points] Find the absolute maximum value of g on [−4, 6]. Justify your answer.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 92: Tabular Rates & Riemann Sum Approximations

★★★★★ VERY IMPORTANT

9 Points

Water is pumped into a municipal storage tank at a rate of R(t) gallons per hour, where t is measured in hours for $0 \le t \le 12$. Values of R(t) are recorded in the table below at selected times:

t (hours): 0, 3, 5, 8, 12
R(t) (gal/hr): 4, 7, 9, 12, 15

(a) [3 Points] Use a trapezoidal sum with the four subintervals indicated by the table to approximate the total volume of water pumped into the tank from $t = 0$ to $t = 12$.

(b) [2 Points] Explain the meaning of $(1/12)\int_0^{12} R(t)\,dt$ in the context of the problem, including appropriate units of measure.

(c) [2 Points] Estimate $R'(6)$ using the data from the table. Show the computations that lead to your answer and include units.

(d) [2 Points] Given that R(t) is strictly increasing on [0, 12], is a Right Riemann sum an overestimate or an underestimate of $\int_0^{12} R(t)\,dt$? Compute this Right Riemann sum.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 93: Fundamental Theorem of Calculus with Composite Bounds

★★★★★ VERY IMPORTANT

9 Points

Let H be the function defined by $H(x) = \int_{1}^{2x} \sqrt{(t^2 + 5)}\, dt$ for all real numbers x.

(a) [2 Points] Evaluate H(1/2). Show your reasoning.

(b) [3 Points] Find H'(x) and evaluate H'(2).

(c) [2 Points] Write an equation of the line tangent to the graph of H at x = 1/2.

(d) [2 Points] Find H''(x) and determine whether the graph of H is concave up or concave down at x = 2.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 94: Definite Integration with Substitution & Particle Motion

★★★★★ VERY IMPORTANT

9 Points

A particle moves along a horizontal coordinate axis so that its velocity at time t (in seconds) is given by $v(t) = t / \sqrt{(t^2 + 9)}$ for $0 \le t \le 4$. At time $t = 0$, the particle is at position $s(0) = 5$ meters.

(a) [2 Points] Find the acceleration a(t) of the particle at time t.

(b) [3 Points] Use the method of substitution to evaluate the displacement $\int_0^4 v(t)\,dt$. Show all steps including the transformed bounds.

(c) [2 Points] Find the position s(4) of the particle at time $t = 4$.

(d) [2 Points] Find the average velocity of the particle over the time interval $[0, 4]$.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 95: Integration by Parts & Improper Integrals (BC Special)

★★★★★ VERY IMPORTANT

9 Points

Consider the function $f(x) = x^2 e^{-x}$ defined for all $x \geq 0$.

(a) [4 Points] Evaluate the indefinite integral $\int x^2 e^{-x}\,dx$ using Integration by Parts or Tabular Integration. Show each differentiation and antiderivative step.

(b) [3 Points] Evaluate the definite integral $\int_0^2 x^2 e^{-x}\,dx$ in exact form.

(c) [2 Points] Evaluate the improper integral $\int_0^\infty x^2 e^{-x}\,dx$ using limits, or show that it diverges.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 96: Partial Fractions & Initial Value Differential Problem (BC Special)

★★★★★ VERY IMPORTANT

9 Points

Consider the differential equation $dy/dx = (x - 5) / (x^2 - x - 6)$ for $x > 3$, with the initial condition $y(4) = \ln 2$.

(a) [3 Points] Find the partial fraction decomposition of $(x - 5) / (x^2 - x - 6)$.

(b) [4 Points] Solve the differential equation to find $y(x)$ explicitly in terms of x for $x > 3$.

(c) [2 Points] Determine $\lim_{x \to \infty} y(x)$.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 97: Improper Integrals Analysis & Parameter Investigation (BC Special)

★★★★★ VERY IMPORTANT

9 Points

Consider the improper integral $I(p) = \int_1^\infty (1 / x^p)\, dx$, where p is a positive real constant.

(a) [3 Points] Evaluate I(2) using the limit definition of an improper integral. State whether it converges or diverges.

(b) [3 Points] Evaluate I(1) using the limit definition of an improper integral. State whether it converges or diverges.

(c) [3 Points] For what values of p does the improper integral I(p) converge? Prove your result by evaluating $\lim_{b\to\infty} \int_1^b$ x^(−p) dx for all p > 0 with p ≠ 1.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 98: Definite Integrals: Symmetry, Properties & Piecewise Analysis

★★★★★ VERY IMPORTANT

9 Points

Let f be a continuous function on $[-6, 6]$ such that $\int_{0}^{6} f(x)\,dx = 10$.

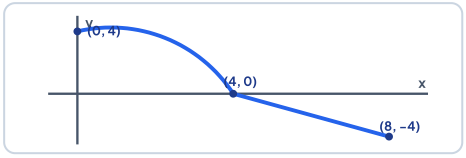


FIGURE FRQ-8: GRAPH OF F(X) ON [0, 8] (QUARTER CIRCLE & LINE SEGMENT)

(a) [2 Points] If f is an even function, evaluate $\int_{-6}^{6} f(x)\,dx$. Justify your answer.

(b) [2 Points] If f is an odd function, evaluate $\int_{-6}^{6} (3f(x) + 4)\,dx$. Justify your answer.

(c) [3 Points] Let $g(x) = f(2x)$. Evaluate $\int_{0}^{3} g(x)\,dx$. Show the substitution that leads to your answer.

(d) [2 Points] Given that $\int_{2}^{6} f(x)\,dx = 7$, find the value of $\int_{0}^{2} f(x)\,dx$.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 99: Accumulation Function with Area & Analytic Pieces

★★★★★ VERY IMPORTANT

9 Points

The continuous function f is defined on [0, 8] by: for $0 \le x \le 4$, $f(x) = \sqrt{(16 - x^2)}$ (the graph is a quarter circle of radius 4 in the first quadrant), and for $4 < x \le 8$, $f(x) = 4 - x$. Let $g(x) = \int_0^x f(t)\,dt$.

(a) [2 Points] Find g(4) and g(8). Show the geometric reasoning.

(b) [2 Points] Find all critical points of g in the open interval (0, 8).

(c) [3 Points] Find the absolute maximum value of g on [0, 8]. Justify your answer.

(d) [2 Points] Write an equation for the line tangent to the graph of g at x = 4.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 100: Comprehensive Mixed Integration Challenge (BC Culmination)

★★★★★ VERY IMPORTANT

9 Points

Consider the functions $f(x) = 1 / (x^2 + 4x + 8)$ and $g(x) = (2x + 4) / (x^2 + 4x + 8)$ defined for all real numbers x.

(a) [3 Points] Use completing the square to find the general antiderivative $\int f(x)\,dx$.

(b) [3 Points] Evaluate the improper integral $\int_{-2}^{\infty} f(x)\,dx$ using proper limit notation.

(c) [3 Points] Use substitution to evaluate the definite integral $\int_{-2}^{0} g(x)\,dx$ in exact logarithmic form.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

End of Question Paper

You have reached the end of Questions 1 through 100.

Before you continue: Make sure you have genuinely attempted every question. The pages that follow contain the complete answer key, fully worked multiple choice solutions with analysis of every incorrect option, and model free response solutions with AP-style scoring rubrics.

What Comes Next

- **Answer Key:** Letter answers for Questions 1–90 with priority star ratings.
- **Multiple Choice Solutions:** Step-by-step reasoning for Questions 1–90 plus an explanation of why each distractor fails.
- **Free Response Solutions:** Complete model answers for Questions 91–100 with point-by-point scoring criteria and rubrics.
- **Appendix:** Essential Unit 6 Rules & Procedures formulas and theorems.

Multiple Choice Answer Key

Letter answers with priority star ratings. Questions 91–100 are free response and are scored using the rubrics that follow.

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
C ★★	D ★★★	C ★★★★★	C ★★★	A ★★★★★	D ★★★★★	A ★★★★★	D ★★★	A ★★★	D ★★★★★
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
A ★★★★★	C ★★★★★	B ★★★	B ★★★★★	A ★★★	C ★★★★★	A ★★★★★	A ★★★★★	A ★★★	B ★★★★★
Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
C ★★★	B ★★★★★	A ★★★	D ★★★★★	A ★★★★★	C ★★★★★	D ★★★	B ★★★★★	B ★★★★★	A ★★★★★
Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
B ★★★★★	C ★★★★★	A ★★★★★	B ★★★★★	A ★★★★★	B ★★★★★	D ★★★	C ★★★★★	A ★★★★	D ★★★★★
Q41	**Q42**	**Q43**	**Q44**	**Q45**	**Q46**	**Q47**	**Q48**	**Q49**	**Q50**
C ★★★★★	B ★★★★★	D ★★★★	A ★★★★★	C ★★★★★	C ★★★★	A ★★★★★	B ★★★	A ★★★★★	A ★★★★
Q51	**Q52**	**Q53**	**Q54**	**Q55**	**Q56**	**Q57**	**Q58**	**Q59**	**Q60**
C ★★★	B ★★	D ★★★★★	A ★★★★★	C ★★★★	B ★★★	D ★★★	B ★★	A ★★★	C ★★★
Q61	**Q62**	**Q63**	**Q64**	**Q65**	**Q66**	**Q67**	**Q68**	**Q69**	**Q70**
A ★★★★★	D ★★★★★	B ★★★★★	C ★★★★	B ★★★★★	C ★★★★	A ★★★★★	D ★★★★★	B ★★★★★	C ★★★★
Q71	**Q72**	**Q73**	**Q74**	**Q75**	**Q76**	**Q77**	**Q78**	**Q79**	**Q80**
C ★★★★★	D ★★★★	B ★★★★★	A ★★★★★	D ★★★★★	B ★★★★★	C ★★★★	D ★★★★★	A ★★★★★	B ★★★★★
Q81	**Q82**	**Q83**	**Q84**	**Q85**	**Q86**	**Q87**	**Q88**	**Q89**	**Q90**
B ★★★★★	D ★★★★★	A ★★★★★	C ★★★★	A ★★★★★	B ★★★★★	D ★★★★★	C ★★★★★	A ★★★★★	B ★★★★★

Scoring Guide: Each multiple choice question is worth 1 point (90 total). Each free response question is worth 9 points (90 total). Maximum raw score: 180 points.

Question 01 Analysis

★★ IMPORTANT

Correct Answer: (C) 14

STEP-BY-STEP SOLUTION

- The total amount of water leaked is the definite integral of the rate of leakage: $\int_1^4 3\sqrt{t}\,dt = \int_1^4 3t^{(1/2)}\,dt = [\,3\cdot(2/3)t^{(3/2)}\,]_1^4 = [\,2t^{(3/2)}\,]_1^4 = 2(4^{(3/2)} - 1^{(3/2)}) = 2(8-1) = 14$ gallons.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Drops the constant multiplier 2 from $3\cdot(2/3)$ and evaluates only $(8-1) = 7$.

✗ **(B)** Evaluates the integrand without antidifferentiating: $3\sqrt{4} + 3\sqrt{1} = 6 + 3 = 9$ or miscomputes $3\cdot 7 = 21$.

✗ **(D)** Computes $2\cdot 8 + 2\cdot 1 = 18$ instead of subtracting the lower bound evaluation.

Question 02 Analysis

★★★ MANDATORY

Correct Answer: (D) 46

STEP-BY-STEP SOLUTION

- With 3 equal subintervals on $[0, 6]$, $\Delta x = (6-0)/3 = 2$.
- The subintervals are $[0, 2]$, $[2, 4]$, and $[4, 6]$.
- The left endpoints are $x = 0, 2, 4$.
- The function values are $f(0) = 1$, $f(2) = 5$, and $f(4) = 17$.
- The Left Riemann sum is $\Delta x[f(0) + f(2) + f(4)] = 2(1 + 5 + 17) = 2(23) = 46$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to multiply the sum of the heights by the subinterval width $\Delta x = 2$.

✗ **(B)** Averages the left and right Riemann sums incorrectly.

✗ **(C)** Computes the Right Riemann sum instead: $2[f(2) + f(4) + f(6)] = 2(5 + 17 + 37) = 118$.

Question 03 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 23

STEP-BY-STEP SOLUTION

- The subintervals have widths: $\Delta x_1 = 3 - 1 = 2$, $\Delta x_2 = 4 - 3 = 1$, $\Delta x_3 = 7 - 4 = 3$.
- Using right endpoints: $R = \Delta x_1 \cdot f(3) + \Delta x_2 \cdot f(4) + \Delta x_3 \cdot f(7) = 2(6) + 1(5) + 3(2) = 12 + 5 + 6 = 23$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes the Left Riemann sum instead: $2(4) + 1(6) + 3(5) = 8 + 6 + 15 = 29$.

✗ **(B)** Assumes equal subinterval widths of 1 for all terms: $6 + 5 + 6 = 17$.

✗ **(D)** Sums all function values $(4 + 6 + 5 + 2 = 17)$ and multiplies by 2.

Question 04 Analysis

★★★ MANDATORY

Correct Answer: (C) 28

STEP-BY-STEP SOLUTION

- With n = 2 on [0, 4], Δx = (4 - 0)/2 = 2.
- The subintervals are [0, 2] and [2, 4].
- The midpoints are x = 1 and x = 3.
- Evaluating f(x) = 2x + 3: f(1) = 5 and f(3) = 9.
- The midpoint Riemann sum is Δx[f(1) + f(3)] = 2(5 + 9) = 2(14) = 28.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to multiply the sum of the midpoint heights by the step width Δx = 2.
✗ **(B)** Uses right endpoints (x = 2, 4) giving 2(7 + 11) = 36 or averages incorrectly.
✗ **(D)** Averages left and right sums with improper arithmetic.

Question 05 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 161

STEP-BY-STEP SOLUTION

- The trapezoidal sum with subintervals [0, 2], [2, 5], and [5, 8] is: T = (2 - 0)·[(10 + 16)/2] + (5 - 2)·[(16 + 22)/2] + (8 - 5)·[(22 + 30)/2] = 2(13) + 3(19) + 3(26) = 26 + 57 + 78 = 161 barrels.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Mistakenly assumes equal subinterval widths of 2 throughout or makes an arithmetic slip.
✗ **(C)** Uses right-endpoint Riemann sum: 2(16) + 3(22) + 3(30) = 32 + 66 + 90 = 188.
✗ **(D)** Uses left-endpoint Riemann sum: 2(10) + 3(16) + 3(22) = 20 + 48 + 66 = 134.

Question 06 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) Left Riemann sum

STEP-BY-STEP SOLUTION

- For any strictly increasing function, on every subinterval [x_{k-1}, x_k], the minimum value occurs at the left endpoint x_{k-1}.
- Consequently, each approximating rectangle lies entirely below the curve, so the Left Riemann sum is guaranteed to be strictly less than the exact integral.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** The Right Riemann sum uses the right endpoint, which is the maximum value on each subinterval, resulting in an overestimate.
✗ **(B)** The Midpoint Riemann sum's over/underestimation depends on concavity (f"), not solely on whether f is increasing.
✗ **(C)** The Trapezoidal sum's over/underestimation depends on concavity (underestimates when concave down, overestimates when concave up).

Question 07 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $T > \int_1^5 f(x)\,dx$

STEP-BY-STEP SOLUTION

- Since $f''(x) > 0$ on $[1, 5]$, the graph of f is strictly concave up.
- For any concave up function, the secant line segment connecting $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$ lies strictly above the curve $y = f(x)$.
- Therefore, the area of each trapezoid exceeds the true area under the curve, meaning $T > \int_1^5 f(x)\,dx$ (an overestimate).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Underestimate occurs when $f''(x) < 0$ (concave down), not when concave up.
✗ **(C)** The trapezoidal sum equals the integral only if f is linear ($f''(x) = 0$).
✗ **(D)** The trapezoidal error sign depends entirely on concavity (f''), independent of whether f is increasing or decreasing.

Question 08 Analysis

★★★ MANDATORY

Correct Answer: (D) 66 gallons

STEP-BY-STEP SOLUTION

- The net rate of change of volume is $V'(t) = E(t) - L(t) = 4t - 2t = 2t$.
- By the Net Change Theorem: $V(4) = V(0) + \int_0^4 (E(t) - L(t))\,dt = 50 + \int_0^4 2t\,dt = 50 + [t^2]_0^4 = 50 + 16 = 66$ gallons.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes only the net accumulation $\int_0^4 2t\,dt = 16$ and forgets to add the initial volume $V(0) = 50$.
✗ **(B)** Adds both rates instead of subtracting: $50 + \int_0^4 6t\,dt = 50 + 48 = 98$ or misadds.
✗ **(C)** Subtracts the accumulation from the initial volume: $50 - 16 = 34$ gallons.

Question 09 Analysis

★★★ MANDATORY

Correct Answer: (A) 11/6

STEP-BY-STEP SOLUTION

- For $[1, 7]$ with $n = 3$, $\Delta x = (7 - 1)/3 = 2$.
- The subintervals are $[1, 3]$, $[3, 5]$, and $[5, 7]$.
- The midpoints are $x = 2, 4, 6$.
- The function values are $f(2) = 1/2$, $f(4) = 1/4$, and $f(6) = 1/6$.
- Midpoint sum $= \Delta x[f(2) + f(4) + f(6)] = 2(1/2 + 1/4 + 1/6) = 2(6/12 + 3/12 + 2/12) = 2(11/12) = 11/6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to multiply the sum of heights 11/12 by the subinterval width $\Delta x = 2$.
✗ **(C)** Uses left endpoints $x = 1, 3, 5$ giving $2(1 + 1/3 + 1/5)$ or makes an algebraic error.
✗ **(D)** Rounds $11/6 \approx 1.833$ up to 2 without exact fraction simplification.

Question 10 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $\pi(\sqrt{2}+1)/4$

STEP-BY-STEP SOLUTION

- With $n = 4$ on $[0, \pi]$, $\Delta x = \pi/4$.
- The grid points are $x_0 = 0, x_1 = \pi/4, x_2 = \pi/2, x_3 = 3\pi/4, x_4 = \pi$.
- By the Trapezoidal Rule: $T = (\Delta x/2)[f(0) + 2f(\pi/4) + 2f(\pi/2) + 2f(3\pi/4) + f(\pi)] = (\pi/8)[0 + 2(\sqrt{2}/2) + 2(1) + 2(\sqrt{2}/2) + 0] = (\pi/8)[\sqrt{2} + 2 + \sqrt{2}] = (\pi/8)[2\sqrt{2} + 2] = (2\pi/8)[\sqrt{2} + 1] = \pi(\sqrt{2} + 1)/4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Counts the midpoint $f(\pi/2)$ with coefficient 4 instead of 2: $(\pi/8)[2\sqrt{2} + 4] = \pi(\sqrt{2} + 2)/4$.

✗ **(B)** Omits the $f(\pi/2) = 1$ term entirely.

✗ **(C)** Forgets to simplify 2/8 to 1/4 in the factored expression.

Question 11 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $\int_1^4 x^2 dx$

STEP-BY-STEP SOLUTION

- In the Riemann sum $\lim_{n\to\infty} \sum_{k=1}^{n} f(x_k)\Delta x$, the subinterval width is $\Delta x = 3/n$ and the grid points are $x_k = 1 + k(3/n) = 1 + 3k/n$.
- Here $a = 1$ and the upper limit is $b = 1 + 3 = 4$.
- The function evaluated at x_k is $f(x_k) = (x_k)^2$, so $f(x) = x^2$.
- Thus, the expression equals $\int_1^4 x^2\, dx$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sets the lower limit to 0 instead of identifying the base offset $a = 1$.

✗ **(C)** Doubles the substitution by keeping the $(1 + 3x)$ form while also extending the limits of integration.

✗ **(D)** Includes an extra factor of 3 outside the integral that was already absorbed by $\Delta x = 3/n$.

Question 12 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1

STEP-BY-STEP SOLUTION

- This expression is a right Riemann sum for $f(x) = \cos x$ on $[0, \pi/2]$, where $\Delta x = \pi/(2n)$ and $x_k = 0 + k(\pi/(2n)) = \pi k/(2n)$.
- Taking the limit as $n\to\infty$ gives the definite integral: $\int_0^{\pi/2} \cos x\, dx = [\sin x]_0^{(\pi/2)} = \sin(\pi/2) - \sin(0) = 1 - 0 = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes the definite integral of cosine over a full quarter period cancels to zero.

✗ **(B)** Multiplies the upper limit $\pi/2$ by the average value of cosine incorrectly.

✗ **(D)** Evaluates the integral across $[-\pi/2, \pi/2]$ instead of $[0, \pi/2]$.

Question 13 Analysis

★★★ MANDATORY

Correct Answer: (B) 2/3

STEP-BY-STEP SOLUTION

- Rewrite the expression as $\lim_{n\to\infty} \sum_{k=1}^{n} \sqrt{(k/n)} \cdot (1/n)$.
- This is the Riemann sum for $f(x) = \sqrt{x}$ on $[0, 1]$ with $\Delta x = 1/n$ and $x_k = k/n$.
- The limit equals: $\int_0^1 x^{(1/2)}\,dx = [(2/3)x^{(3/2)}]_0^1 = 2/3(1 - 0) = 2/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses $\sqrt{x}$ with the linear function x, yielding $[x^2/2]_0^1 = 1/2$.

✗ **(C)** Integrates with power 1/3 or forgets the reciprocal in the power rule $1/(1/2 + 1) = 2/3$.

✗ **(D)** Inverts the coefficient to 3/2 instead of 2/3.

Question 14 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\int_2^7 x^3 dx$

STEP-BY-STEP SOLUTION

- Here $\Delta x = 5/n$ and the grid points are $x_k = 2 + 5k/n$.
- The lower limit is $a = 2$ and the upper limit is $b = 2 + 5 = 7$.
- The function evaluated at x_k is $f(x_k) = (x_k)^3$, so $f(x) = x^3$.
- Therefore, the limit represents $\int_2^7 x^3\,dx$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** If interpreted on $[0, 5]$, $\Delta x = 5/n$ would correspond to $x_k = 5k/n$ and the integrand would be $(2 + x)^3$, but with dx (not $5\,dx$). While $\int_0^5 (2 + x)^3\,dx$ has the same numerical value, the standard translation with $x = 2 + 5k/n$ is $\int_2^7 x^3\,dx$.

✗ **(C)** Retains an extraneous constant factor 5 outside the integral.

✗ **(D)** Ignores the base shift 2 entirely.

Question 15 Analysis

★★★ MANDATORY

Correct Answer: (A) 22

STEP-BY-STEP SOLUTION

- For any uniform partition of an interval, the trapezoidal sum T is the arithmetic mean of the Left Riemann sum L and the Right Riemann sum R: $T = (L + R) / 2 = (18 + 26) / 2 = 44 / 2 = 22$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Subtracts the differences: $26 - 18 = 8$, then adjusts $18 + 2 = 20$.

✗ **(C)** Averages 22 and 26 incorrectly.

✗ **(D)** Sums $L + R = 44$ without dividing by 2.

Question 16 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $\int_0^2 e^x\,dx$

STEP-BY-STEP SOLUTION

- Let $x_k = 2k/n$ and $\Delta x = 2/n$.
- The lower bound is $a = 0$ and the upper bound is $b = 0 + 2 = 2$.
- The function evaluated at x_k is $f(x_k) = e^{x_k}$, so $f(x) = e^x$.
- Therefore, the limit represents $\int_0^2 e^x\,dx$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** If viewed on $[0, 1]$, Δx would be $1/n$, so the factor $2/n$ would leave an extra factor of 2, making it $2\int_0^1 e^{2x}\,dx$, not $\int_0^1 e^{2x}\,dx$.

✗ **(B)** Includes an extra factor of 2 outside the integral.

✗ **(D)** Starts the lower limit at 1 instead of 0.

Question 17 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) T < I < M

STEP-BY-STEP SOLUTION

- When $f''(x) < 0$ on $[a, b]$, the graph of f is concave down.
- For any concave down curve: (1) Secant segments connecting interval endpoints lie strictly below the curve, so the Trapezoidal sum underestimates the true area ($T < I$). (2) Tangent lines drawn at the midpoints lie strictly above the curve, so the Midpoint sum overestimates the true area ($I < M$).
- Combining these gives $T < I < M$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** This order applies when f is concave up ($f''(x) > 0$), where T overestimates and M underestimates.

✗ **(C)** Trapezoidal sum cannot overestimate when the function is concave down.

✗ **(D)** Midpoint sum does not underestimate when f is concave down.

Question 18 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) π/4

STEP-BY-STEP SOLUTION

- With $\Delta x = 1/n$ and $x_k = k/n$ on $[0, 1]$, the limit is equivalent to the definite integral $\int_0^1 [1 / (1 + x^2)]\,dx$.
- Antidifferentiating gives $[\arctan x]_0^1 = \arctan(1) - \arctan(0) = \pi/4 - 0 = \pi/4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Evaluates $\arctan(1)$ as $\pi/2$ instead of $\pi/4$.

✗ **(C)** Mistakes $1/(1+x^2)$ for $1/(1+x)$, which would integrate to $\ln(1+x)|_0^1 = \ln 2$.

✗ **(D)** Neglects the inverse tangent antiderivative and simply evaluates $1/(1+0) = 1$.

Question 19 Analysis

★★★ MANDATORY

Correct Answer: (A) 51 meters

STEP-BY-STEP SOLUTION

- The subintervals all have equal width $\Delta t = 3$.
- Using the Trapezoidal Rule: Distance $\approx (\Delta t / 2)[v(0) + 2v(3) + 2v(6) + v(9)] = (3/2)[2 + 2(5) + 2(7) + 8] = (3/2)[2 + 10 + 14 + 8] = (3/2)[34] = 3 \cdot 17 = 51$ meters.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses $\Delta t = 3$ but omits the division by 2: $3 \cdot (2 + 5 + 7 + 8) = 66$ or $2 \cdot 34 = 68$.

✗ **(C)** Omits the step width $\Delta t = 3$ and reports only the sum of terms $(34/2) = 17$ or 34.

✗ **(D)** Uses a Left Riemann sum: $3(2 + 5 + 7) = 3(14) = 42$ or right Riemann sum.

Question 20 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\lim_{n\to\infty} \sum_{k=1}^{n} \ln(3 + 2k/n) \cdot (2/n)$

STEP-BY-STEP SOLUTION

- For the interval $[3, 5]$, the subinterval width is $\Delta x = (5 - 3)/n = 2/n$.
- The right endpoints are $x_k = a + k\Delta x = 3 + k(2/n) = 3 + 2k/n$.
- Evaluating the integrand $f(x) = \ln(x)$ at x_k gives $\ln(3 + 2k/n)$.
- Therefore, the Riemann sum limit is $\lim_{n\to\infty} \sum_{k=1}^{n} \ln(3 + 2k/n) \cdot (2/n)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses grid points $x_k = 3 + k/n$, which corresponds to an interval of length 1 (width $1/n$) instead of $2/n$.

✗ **(C)** Omits the lower limit $a = 3$, corresponding to $[0, 2]$ instead of $[3, 5]$.

✗ **(D)** Uses $\Delta x = 1/n$ instead of the required $(5 - 3)/n = 2/n$.

Question 21 Analysis

★★★ MANDATORY

Correct Answer: (C) 3

STEP-BY-STEP SOLUTION

- By the Fundamental Theorem of Calculus Part 1, $g'(x) = d/dx\left[\int_{2}^{2x} \sqrt{(t^3 + 1)}\, dt\right] = \sqrt{(x^3 + 1)}$.
- Evaluating at $x = 2$ gives $g'(2) = \sqrt{(2^3 + 1)} = \sqrt{(8 + 1)} = \sqrt{9} = 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses $g'(2)$ with $g(2) = \int_{2}^{2} \sqrt{(t^3 + 1)}\, dt = 0$.

✗ **(B)** Forgets to take the square root of $8 + 1 = 9$.

✗ **(D)** Differentiates the radicand using the power rule $3t^2$ and evaluates at 2.

Question 22 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $2x\cos(x^2)$

STEP-BY-STEP SOLUTION

- By the Fundamental Theorem of Calculus combined with the Chain Rule: $d/dx\left[\int_0^{u(x)}\cos(t)\,dt\right] = \cos(u(x))\cdot u'(x)$.
- With $u(x) = x^2$, $u'(x) = 2x$, so the derivative is $\cos(x^2)\cdot 2x = 2x\cos(x^2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to apply the chain rule factor $2x$ from the upper limit.

✗ **(C)** Antidifferentiates or differentiates the cosine function to -sine.

✗ **(D)** Integrates cos(t) to sin(t) first and forgets to differentiate.

Question 23 Analysis

★★★★ MANDATORY

Correct Answer: (A) $-e^{x^2}$

STEP-BY-STEP SOLUTION

- First reverse the limits of integration using the property $\int_{x^5}^{} e^{t^2}\,dt = -\int_{5x}^{} e^{t^2}\,dt$.
- Now applying the Fundamental Theorem of Calculus: $d/dx\left[-\int_{5x}^{} e^{t^2}\,dt\right] = -e^{x^2}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Drops the negative sign that arises from having the variable x in the lower bound.

✗ **(C)** Applies the chain rule derivative $2x$ to the exponent unnecessarily while dropping the sign.

✗ **(D)** Introduces an unnecessary inner derivative $2x$ from the exponent.

Question 24 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 3

STEP-BY-STEP SOLUTION

- By the Fundamental Theorem of Calculus and the Chain Rule: $h'(x) = \ln(3x)\cdot d/dx[3x] = 3\ln(3x)$.
- Evaluating at $x = e/3$: $h'(e/3) = 3\ln(3\cdot(e/3)) = 3\ln(e) = 3(1) = 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the chain rule factor 3: $\ln(e) = 1$.

✗ **(B)** Confuses $\ln(e)$ with e.

✗ **(C)** Multiplies the chain factor 3 by e instead of $\ln(e)$.

Question 25 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $(3x^2/(x^3+1)) - (2/(2x+1))$

STEP-BY-STEP SOLUTION

- Split the integral at a constant c: $\int_{2x}^{x^3} = \int_{c}^{x^3} - \int_{c}^{2x}$.
- By the Leibniz rule (FTC + Chain Rule): $d/dx = f(x^3) \cdot d/dx[x^3] - f(2x) \cdot d/dx[2x] = [1/(x^3+1)] \cdot 3x^2 - [1/(2x+1)] \cdot 2 = (3x^2/(x^3+1)) - (2/(2x+1))$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits both chain rule factors $3x^2$ and 2.

✗ **(C)** Adds the two boundary contributions instead of subtracting the lower bound derivative.

✗ **(D)** Combines numerators and denominators in an invalid algebraic shortcut.

Question 26 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $y = 3x - 9$

STEP-BY-STEP SOLUTION

- First, find the y-coordinate: $g(3) = \int_{3}^{3} (t^2 - 2t)\,dt = 0$, so the point of tangency is (3, 0).
- Next, find the slope: by FTC 1, $g'(x) = x^2 - 2x$.
- At $x = 3$, $g'(3) = 3^2 - 2(3) = 9 - 6 = 3$.
- Using point-slope form: $y - 0 = 3(x - 3) \Rightarrow y = 3x - 9$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses slope 3 but ignores the horizontal translation $(x - 3)$.

✗ **(B)** Sign error in expanding $3(x - 3) = 3x + 9$.

✗ **(D)** Leaves the derivative in terms of x rather than evaluating the numerical slope at $x = 3$.

Question 27 Analysis

★★★ MANDATORY

Correct Answer: (D) 1

STEP-BY-STEP SOLUTION

- By FTC 1, $f'(x) = x \sin(x)$.
- To find the second derivative, apply the Product Rule to $f'(x)$: $f''(x) = d/dx\,[x \sin x] = (1)\cdot\sin(x) + x\cdot\cos(x) = \sin(x) + x\cos(x)$.
- Evaluating at $x = \pi/2$ gives: $f''(\pi/2) = \sin(\pi/2) + (\pi/2)\cos(\pi/2) = 1 + (\pi/2)(0) = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates $f'(0)$ instead of $f''(\pi/2)$.

✗ **(B)** Retains the term $(\pi/2)\cos(\pi/2)$ and mistakenly assumes $\cos(\pi/2) = 1$.

✗ **(C)** Sign error on $\sin(\pi/2)$.

Question 28 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) x = 2

STEP-BY-STEP SOLUTION

- By FTC 1, $F'(x) = x^2 - 4$.
- Setting $F'(x) = 0$ gives $x^2 - 4 = 0 \Rightarrow x = 2$ (since the domain is $x > 0$).
- For $0 < x < 2$, $F'(x) < 0$ (F is decreasing), and for $x > 2$, $F'(x) > 0$ (F is increasing).
- Since F' changes from negative to positive at $x = 2$, F has a relative minimum at $x = 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Tests the lower integration limit $x = 1$ instead of critical points of F.
✗ **(C)** Sets the linear factor $x - 4 = 0$.
✗ **(D)** Ignores the domain restriction $x > 0$.

Question 29 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\cos x \cdot e^{\sin^2 x}$

STEP-BY-STEP SOLUTION

- Applying FTC 1 with the Chain Rule: let $u = \sin x$.
- Then $d/dx\left[\int_0^u e^{t^2}\,dt\right] = e^{u^2} \cdot du/dx = e^{(\sin x)^2} \cdot \cos x = \cos x \cdot e^{\sin^2 x}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to multiply by the derivative of the upper limit $du/dx = \cos x$.
✗ **(C)** Introduces an extraneous negative sign.
✗ **(D)** Differentiates inside the exponent using the chain rule on $\sin^2 x$.

Question 30 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 1

STEP-BY-STEP SOLUTION

- As $x \to 0$, the numerator approaches $\int_0^0 \cos(t^2)\,dt = 0$, and the denominator approaches 0, which is an indeterminate form 0/0.
- Applying L'Hôpital's Rule: differentiate the numerator using FTC 1: $d/dx\left[\int_0^x \cos(t^2)\,dt\right] = \cos(x^2)$.
- Differentiate the denominator: $d/dx[x] = 1$.
- The limit becomes: $\lim_{x \to 0} [\cos(x^2) / 1] = \cos(0) / 1 = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Assumes that the limit of an integral evaluating to 0 at 0 must be 0.
✗ **(C)** Confuses $\cos(0)$ with π or $\pi/2$.
✗ **(D)** Concludes the limit does not exist without applying L'Hôpital's Rule.

Question 31 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 0

STEP-BY-STEP SOLUTION

- The value g(4) is the net signed area from t = 0 to t = 4: $g(4) = \int_0^4 f(t)\,dt = \int_0^2 f(t)\,dt + \int_2^4 f(t)\,dt$.
- On [0, 2], the region is a triangle above the t-axis with base 2 and height 4: Area = (1/2)(2)(4) = 4.
- On [2, 4], the region is a triangle below the t-axis with base 2 and height -4: Area = -(1/2)(2)(4) = -4.
- Thus, g(4) = 4 + (-4) = 0.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds the absolute areas: 4 + 4 = 8, ignoring that the region on [2, 4] is below the t-axis.
✗ **(C)** Evaluates only the positive triangle on [0, 2] and neglects the interval [2, 4].
✗ **(D)** Reports only the negative triangle area on [2, 4].

Question 32 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) –4

STEP-BY-STEP SOLUTION

- By properties of definite integrals: $g(-2) = \int_0^{-2} f(t)\,dt = -\int_{-2}^0 f(t)\,dt$.
- On [-2, 0], f is above the t-axis and forms a triangle with base 2 and height 4, so $\int_{-2}^0 f(t)\,dt = (1/2)(2)(4) = 4$.
- Therefore, g(-2) = -(4) = -4.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets that integrating backwards from 0 to -2 reverses the sign of the area.
✗ **(B)** Assumes g(-2) = 0 because f(-2) = 0.
✗ **(D)** Doubles the negative area unnecessarily.

Question 33 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) (–2, 2)

STEP-BY-STEP SOLUTION

- By FTC 1, g'(x) = f(x).
- A function g is strictly increasing where its derivative is positive: $g'(x) > 0 \Leftrightarrow f(x) > 0$.
- The graph of f is strictly above the x-axis for all x in (-2, 2).
- Therefore, g is strictly increasing on (-2, 2).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Confuses where f is decreasing with where g is increasing.
✗ **(C)** On (2, 6), f(x) < 0, which means g is decreasing, not increasing.
✗ **(D)** Selects only the portion where f has a positive slope rather than where f itself is positive.

Question 34 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) x = 2

STEP-BY-STEP SOLUTION

- By the Closed Interval Method, test endpoints and critical points (where $g'(x) = f(x) = 0$): Endpoints: $g(-2) = -4$, $g(6) = \int_0^6 f(t)\,dt = 4 - 4 - 4 = -4$.
- Critical point where f changes sign: at $x = 2$, f changes from positive to negative, so $g(2) = 4$ is a local maximum.
- Comparing values: $g(-2) = -4$, $g(2) = 4$, $g(6) = -4$.
- The absolute maximum of g is 4, which occurs at $x = 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** $x = 0$ is where f achieves its maximum, not where the accumulation function g achieves its maximum.

✗ **(C)** At $x = 4$, $g(4) = 0$, which is strictly less than $g(2) = 4$.

✗ **(D)** At $x = 6$, $g(6) = -4$, which is the absolute minimum.

Question 35 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) (0, 4)

STEP-BY-STEP SOLUTION

- The graph of g is concave down where $g''(x) < 0$.
- Since $g'(x) = f(x)$, $g''(x) = f'(x)$.
- Therefore, g is concave down where f is decreasing (slope of f is negative).
- The graph of f decreases along the line segment from (0, 4) to (4, -4), which corresponds to the interval (0, 4).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** On (4, 6), f increases, so g is concave up.

✗ **(C)** On (-2, 0), f increases from 0 to 4, so $f' > 0$ and g is concave up.

✗ **(D)** On (4, 6), the slope of f is positive (+2), so g is concave up.

Question 36 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) x = 0

STEP-BY-STEP SOLUTION

- A point of inflection of g occurs where $g''(x)$ changes sign.
- Since $g''(x) = f'(x)$, this corresponds to where f changes from increasing to decreasing or from decreasing to increasing (i.e., at local extrema of f).
- At $x = 0$, f changes from increasing to decreasing, meaning f' (and hence g'') changes from positive to negative.
- Therefore, $x = 0$ is a point of inflection of g.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** At $x = 2$, $f(x) = 0$ and f is strictly decreasing with constant slope -2; g'' does not change sign.

✗ **(C)** $x = -2$ is a boundary endpoint of the domain.

✗ **(D)** At $x = 1$, f is in the middle of a linear segment with constant negative slope.

Question 37 Analysis

★★★ MANDATORY

Correct Answer: (D) 14

STEP-BY-STEP SOLUTION

- Substitute $x = 4$ directly into the definition of $h(x)$: $h(4) = 5 + \int_{1}^{4} f(t)\,dt$.
- Since $\int_{1}^{4} f(t)\,dt = 9$, we have $h(4) = 5 + 9 = 14$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the initial constant 5: reports only $\int_{1}^{4} f(t)\,dt = 9$.
✗ **(B)** Subtracts 5 from 9: $9 - 5 = 4$.
✗ **(C)** Subtracts 9 from 5: $5 - 9 = -4$.

Question 38 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 2π

STEP-BY-STEP SOLUTION

- The value $g(4) = \int_{0}^{4} f(t)\,dt$ is the area under the semicircle of radius $r = 2$ on $[0, 4]$.
- The area of a semicircle is $(1/2)\pi r^2 = (1/2)\pi(2^2) = (1/2)\pi(4) = 2\pi$.
- Since the semicircle is entirely above the x-axis, $g(4) = 2\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes the area of a full circle $\pi r^2 = 4\pi$ without dividing by 2.
✗ **(B)** Uses radius $r = 1$ giving $(1/2)\pi(1) = \pi/2$ or confuses diameter with radius.
✗ **(D)** Approximates the semicircle by a rectangle or bounding triangle.

Question 39 Analysis

★★★★ MANDATORY

Correct Answer: (A) Increasing and concave down

STEP-BY-STEP SOLUTION

- By FTC 1, $g'(x) = f(x)$.
- Since $f(x) > 0$, $g'(x) > 0$, so g is strictly increasing.
- Furthermore, $g''(x) = f'(x)$.
- Since f is strictly decreasing, $f'(x) < 0$, which means $g''(x) < 0$.
- A negative second derivative implies g is concave down.
- Thus, g is increasing and concave down.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Incorrectly concludes concave up despite f being decreasing.
✗ **(C)** Mistakes a decreasing derivative for a decreasing function.
✗ **(D)** Combines both incorrect attributes.

Question 40 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $(-\infty, \ln 3)$

STEP-BY-STEP SOLUTION

- By FTC 1, $H'(x) = e^x - 3$.
- The function H is strictly decreasing when $H'(x) < 0$.
- Set $e^x - 3 < 0 \Rightarrow e^x < 3 \Rightarrow x < \ln 3$.
- Therefore, H is strictly decreasing on $(-\infty, \ln 3)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the interval where $H'(x) > 0$, which means H is increasing.
✗ **(B)** Solves $x < 3$ without applying the natural logarithm to e^x.
✗ **(C)** Confuses the bound 3 with $\ln 3$.

Question 41 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 18

STEP-BY-STEP SOLUTION

- By interval additivity, $\int_1^3 f(x)\,dx = \int_1^5 f(x)\,dx - \int_3^5 f(x)\,dx = 10 - 4 = 6$.
- By linearity: $\int_1^3 (2f(x) + 3)\,dx = 2\int_1^3 f(x)\,dx + \int_1^3 3\,dx = 2(6) + 3(3 - 1) = 12 + 6 = 18$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to multiply the constant 3 by the interval width $(3 - 1) = 2$, adding only 3: $12 + 3 = 15$.
✗ **(B)** Evaluates only $2\int_1^3 f(x)\,dx = 12$ and drops the constant term entirely.
✗ **(D)** Adds the integrals instead of subtracting: $10 + 4 = 14$, then $2(14) - 7$.

Question 42 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 4

STEP-BY-STEP SOLUTION

- Reversing the limits of integration gives $\int_4^6 f(x)\,dx = -\int_6^4 f(x)\,dx = -3$.
- Then by additivity: $\int_0^6 f(x)\,dx = \int_0^4 f(x)\,dx + \int_4^6 f(x)\,dx = 7 + (-3) = 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Fails to reverse the sign when flipping the limits from 6 to 4: computes $7 + 3 = 10$.
✗ **(C)** Reverses the sign of both integrals: $-7 + 3 = -4$.
✗ **(D)** Multiplies the values $7 \cdot 3 = 21$.

Question 43 Analysis

★★★★ MANDATORY

Correct Answer: (D) 20

STEP-BY-STEP SOLUTION

- Notice that $g(x) = x^5 - 3x^3 + 4\sin x$ is an odd function because $g(-x) = -g(x)$.
- The integral of an odd function over any symmetric interval $[-a, a]$ is zero: $\int_{-2}^{2} (x^5 - 3x^3 + 4\sin x)\,dx = 0$.
- Therefore, the integral reduces to: $\int_{-2}^{2} 5\,dx = 5(2 - (-2)) = 5(4) = 20$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes the entire integrand is odd, incorrectly claiming the integral is 0.
✗ **(B)** Evaluates $5(2) = 10$ without accounting for the lower limit -2.
✗ **(C)** Doubles the result: $20 \cdot 2 = 40$.

Question 44 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 4

STEP-BY-STEP SOLUTION

- Since f is an even function, $\int_{-3}^{3} f(x)\,dx = 2\int_{0}^{3} f(x)\,dx = 2(8) = 16$.
- The integral of the constant term is $\int_{-3}^{3} 2\,dx = 2(3 - (-3)) = 2(6) = 12$.
- Subtracting the two gives: $\int_{-3}^{3} (f(x) - 2)\,dx = 16 - 12 = 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to integrate the constant term -2 and reports only $2(8) = 16$.
✗ **(C)** Subtracts $2 \cdot 3 = 6$ instead of $2 \cdot 6 = 12$: $16 - 6 = 10$.
✗ **(D)** Treats f as an odd function so $\int_{-3}^{3} f(x)\,dx = 0$.

Question 45 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 24

STEP-BY-STEP SOLUTION

- By the Fundamental Theorem of Calculus (Net Change Theorem): $x(3) = x(1) + \int_{1}^{3} v(t)\,dt$.
- Compute the definite integral: $\int_{1}^{3} (3t^2 - 2t)\,dt = [t^3 - t^2]_1^3 = (3^3 - 3^2) - (1^3 - 1^2) = (27 - 9) - (1 - 1) = 18$.
- Therefore, $x(3) = 6 + 18 = 24$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Finds the displacement $\int_{1}^{3} v(t)\,dt = 18$ but forgets to add the initial position $x(1) = 6$.
✗ **(B)** Adds displacement to velocity at $t = 3$: $18 + 12 = 30$.
✗ **(D)** Subtracts the displacement: $18 - 6 = 12$.

Question 46 Analysis

★★★★ MANDATORY

Correct Answer: (C) 4

STEP-BY-STEP SOLUTION

- Split the integral where the absolute value argument changes sign at $x = 2$: $\int_0^4 |x-2|\,dx = \int_0^2 -(x-2)\,dx + \int_2^4 (x-2)\,dx = \int_0^2 (2-x)\,dx + \int_2^4 (x-2)\,dx$.
- Geometrically, this represents two congruent right triangles each with base 2 and height 2: Area $= (1/2)(2)(2) + (1/2)(2)(2) = 2 + 2 = 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Integrates without absolute value: $\int_0^4 (x-2)\,dx = [x^2/2 - 2x]_0^4 = 8 - 8 = 0$.

✗ **(B)** Computes the area of only one triangle.

✗ **(D)** Doubles the base times height without dividing by 2: $2 \cdot 2 + 2 \cdot 2 = 8$.

Question 47 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 8

STEP-BY-STEP SOLUTION

- Split the integral at the boundary $x = 1$: $\int_0^2 f(x)\,dx = \int_0^1 2x\,dx + \int_1^2 3x^2\,dx$.
- Evaluate each piece: $\int_0^1 2x\,dx = [x^2]_0^1 = 1 - 0 = 1$. $\int_1^2 3x^2\,dx = [x^3]_1^2 = 2^3 - 1^3 = 8 - 1 = 7$.
- Summing the two parts gives $1 + 7 = 8$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Computes only the second piece $\int_1^2 3x^2\,dx = 7$.

✗ **(C)** Integrates $3x^2$ over the entire interval $[0, 2]$: $[x^3]_0^2 = 8 + 1 = 9$.

✗ **(D)** Integrates $2x$ over the entire interval $[0, 2]$: $[x^2]_0^2 = 4$, then adds 2.

Question 48 Analysis

★★★ MANDATORY

Correct Answer: (B) 6

STEP-BY-STEP SOLUTION

- The average value of a continuous function f on $[a, b]$ is $(1/(b-a))\int_a^b f(x)\,dx$.
- Here: $f_{avg} = (1/(3-0))\int_0^3 (3x^2 - 2x)\,dx = (1/3)[x^3 - x^2]_0^3 = (1/3)[(27 - 9) - 0] = (1/3)(18) = 6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reports the definite integral value $\int_0^3 (3x^2 - 2x)\,dx = 18$ without dividing by the interval width $(b - a) = 3$.

✗ **(C)** Divides 18 by 2 instead of the interval length 3.

✗ **(D)** Divides by 3 twice: $18 / 6 = 3$.

Question 49 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $(1 + e^{\pi/2}) / 2$

STEP-BY-STEP SOLUTION

- Find the antiderivative of each term: $\int \sec^2 x\,dx = \tan x$, and $\int e^{2x}\,dx = (1/2)e^{2x}$.
- So the antiderivative is $F(x) = \tan x + (1/2)e^{2x}$.
- Evaluating at the limits: At $x = \pi/4$: $\tan(\pi/4) + (1/2)e^{2(\pi/4)} = 1 + (1/2)e^{\pi/2}$.
- At $x = 0$: $\tan(0) + (1/2)e^0 = 0 + 1/2 = 1/2$.
- Subtracting gives: $[1 + (1/2)e^{\pi/2}] - 1/2 = 1/2 + (1/2)e^{\pi/2} = (1 + e^{\pi/2}) / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets the factor 1/2 in the antiderivative of e^{2x}.
✗ **(C)** Neglects $\tan(\pi/4) = 1$ in the upper limit.
✗ **(D)** Multiplies by 2 instead of dividing by 2 when integrating e^{2x}.

Question 50 Analysis

★★★★ MANDATORY

Correct Answer: (A) $e^3 + 1$

STEP-BY-STEP SOLUTION

- Find the antiderivative: $F(x) = 2\ln|x| + x^3$.
- Evaluating at the limits: At $x = e$: $F(e) = 2\ln(e) + e^3 = 2(1) + e^3 = 2 + e^3$.
- At $x = 1$: $F(1) = 2\ln(1) + 1^3 = 0 + 1 = 1$.
- Subtracting gives: $F(e) - F(1) = (2 + e^3) - 1 = e^3 + 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits subtracting $F(1) = 1$ and reports $F(e) = e^3 + 2$.
✗ **(C)** Subtracts $2\ln(e)$ instead of adding it in $F(e)$.
✗ **(D)** Multiplies the entire expression by 2.

Question 51 Analysis

★★★ MANDATORY

Correct Answer: (C) $x^2/2 - 4x - 2/x + C$

STEP-BY-STEP SOLUTION

- Divide each term in the numerator by the monomial denominator x^2 before integrating: $(x^3 - 4x^2 + 2)/x^2 = x - 4 + 2x^{-2}$.
- Integrating term by term using the Power Rule: $\int (x - 4 + 2x^{-2})\,dx = x^2/2 - 4x + 2(x^{-1} / (-1)) + C = x^2/2 - 4x - 2/x + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error on the last term: keeps $+2/x$ instead of $-2/x$ from integrating $2x^{-2}$.
✗ **(B)** Incorrectly integrates numerator and denominator separately.
✗ **(D)** Treats x^{-2} as a linear term.

Question 52 Analysis

★★ IMPORTANT

Correct Answer: (B) $5^x / \ln 5 + C$

STEP-BY-STEP SOLUTION

- For any base $a > 0$ with $a \neq 1$, the antiderivative of a^x is $(a^x / \ln a) + C$.
- For $a = 5$, $\int 5^x \, dx = (5^x / \ln 5) + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Multiplies by $\ln 5$ instead of dividing; $5^x \ln 5$ is the derivative of 5^x.
✗ **(C)** Applies the power rule incorrectly to an exponential function.
✗ **(D)** Omits the scaling factor $1 / \ln 5$.

Question 53 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) (1/3) arcsin(3x) + C

STEP-BY-STEP SOLUTION

- Recognize the form for inverse sine: $\int (1 / \sqrt{(1 - u^2)}) \, du = \arcsin(u) + C$.
- Let $u = 3x$, so $du = 3 \, dx$ and $dx = du / 3$.
- Substituting gives: $(1/3) \int (1 / \sqrt{(1 - u^2)}) \, du = (1/3) \arcsin(u) + C = (1/3) \arcsin(3x) + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to account for the chain rule factor 1/3 from $du = 3 \, dx$.
✗ **(B)** Uses coefficient 1/9 instead of 1/3.
✗ **(C)** Uses inverse tangent instead of inverse sine.

Question 54 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) (1/4) arctan(x/4) + C

STEP-BY-STEP SOLUTION

- Recall the standard formula $\int (1 / (x^2 + a^2)) \, dx = (1/a) \arctan(x/a) + C$.
- Here $a^2 = 16$, so $a = 4$.
- Applying the formula gives $(1/4) \arctan(x/4) + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits the leading coefficient $1/a = 1/4$ outside the arctangent function.
✗ **(C)** Uses $a = 16$ instead of $a = \sqrt{16} = 4$.
✗ **(D)** Uses the arcsine function instead of arctangent.

Question 55 Analysis

★★★★ MANDATORY

Correct Answer: (C) $\tan x - x + C$

STEP-BY-STEP SOLUTION

- Use the Pythagorean identity $\tan^2 x = \sec^2 x - 1$: $\int \tan^2 x\,dx = \int (\sec^2 x - 1)\,dx$.
- Since the antiderivative of $\sec^2 x$ is $\tan x$, this gives $\tan x - x + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Incorrectly applies the power rule to $\tan x$ without having $\sec^2 x\,dx$ present.
✗ **(B)** Replaces $\tan^2 x$ with its derivative $\sec^2 x$.
✗ **(D)** Sign error: adds x instead of subtracting x.

Question 56 Analysis

★★★ MANDATORY

Correct Answer: (B) $y = x^4 + x^2 - x + 4$

STEP-BY-STEP SOLUTION

- Find the general antiderivative: $y = \int (4x^3 + 2x - 1)\,dx = x^4 + x^2 - x + C$.
- Use the initial condition $y(1) = 5$: $5 = (1)^4 + (1)^2 - (1) + C \Rightarrow 5 = 1 + C \Rightarrow C = 4$.
- Therefore, $y = x^4 + x^2 - x + 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sets C equal to the initial value 5 without solving for the constant.
✗ **(C)** Differentiates dy/dx to find the second derivative rather than integrating.
✗ **(D)** Makes an arithmetic error: solves $5 = 2 + C$ to get $C = 3$.

Question 57 Analysis

★★★ MANDATORY

Correct Answer: (D) $2x - 6\sqrt{x} + C$

STEP-BY-STEP SOLUTION

- Split the fraction: $(2\sqrt{x} - 3)/\sqrt{x} = 2 - 3x\text{^}(-1/2)$.
- Integrate term by term: $\int (2 - 3x\text{^}(-1/2))\,dx = 2x - 3 \cdot [x\text{^}(1/2) / (1/2)] + C = 2x - 3(2\sqrt{x}) + C = 2x - 6\sqrt{x} + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to divide by the new exponent 1/2, keeping $-3\sqrt{x}$.
✗ **(B)** Multiplies by 1/2 instead of dividing by 1/2: $-3 \cdot (1/2) = -3/2$.
✗ **(C)** Integrates 2 as x instead of $2x$.

Question 58 Analysis

★★ IMPORTANT

Correct Answer: (B) $-(1/4)\csc(4x) + C$

STEP-BY-STEP SOLUTION

- Recall the standard derivative rule: $d/du[\csc u] = -\csc u \cot u$, so $\int \csc u \cot u\, du = -\csc u + C$.
- Here $u = 4x$, so $du = 4\,dx$ and $dx = du/4$.
- Thus: $\int \csc(4x)\cot(4x)\,dx = -(1/4)\csc(4x) + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the essential negative sign from the cosecant antiderivative.
✗ **(C)** Multiplies by 4 instead of dividing by 4.
✗ **(D)** Mistakes cosecant for secant.

Question 59 Analysis

★★★ MANDATORY

Correct Answer: (A) 7/8

STEP-BY-STEP SOLUTION

- Rewrite the integrand with a negative exponent: $3/x^4 = 3x^{-4}$.
- Antidifferentiate: $\int 3x^{-4}\,dx = 3 \cdot [x^{-3}/(-3)] = -x^{-3} = -1/x^3$.
- Evaluate between limits 1 and 2: $[-1/x^3]_1^2 = (-1/2^3) - (-1/1^3) = -1/8 - (-1) = 1 - 1/8 = 7/8$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error in subtracting the lower limit: $(-1/8) - 1 = -9/8$ or $-7/8$.
✗ **(C)** Integrates with power -3 instead of -4.
✗ **(D)** Divides 7/8 by 3 unnecessarily.

Question 60 Analysis

★★★ MANDATORY

Correct Answer: (C) 1

STEP-BY-STEP SOLUTION

- The antiderivative of $\sec x \tan x$ is $\sec x$: $\int_0^{\pi/3} \sec x \tan x\,dx = [\sec x]_0^{(\pi/3)} = \sec(\pi/3) - \sec(0)$.
- Since $\cos(\pi/3) = 1/2$, $\sec(\pi/3) = 2$.
- Since $\cos(0) = 1$, $\sec(0) = 1$.
- Therefore: $2 - 1 = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates $\sec(\pi/3) = 2$ but forgets to subtract $\sec(0) = 1$.
✗ **(B)** Confuses the antiderivative with $\tan x$, evaluating $\tan(\pi/3) - \tan(0) = \sqrt{3} - 0 = \sqrt{3}$.
✗ **(D)** Assumes the trigonometric evaluation cancels to zero.

Question 61 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) (1/12) $(x^2+4)^6+C$

STEP-BY-STEP SOLUTION

- Let $u = x^2 + 4$, so $du = 2x\,dx$ and $x\,dx = du / 2$.
- Substituting yields: $\int x(x^2+4)^5\,dx = (1/2)\int u^5\,du = (1/2)\cdot(u^6/6) + C = (1/12)u^6 + C = (1/12)(x^2+4)^6 + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to divide by 2 from the substitution $x\,dx = du / 2$.

✗ **(C)** Multiplies by 2 instead of dividing by 2.

✗ **(D)** Forgets to increase the power from 5 to 6.

Question 62 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) (1/2) $\int_1^9 \sqrt{u}\,du$

STEP-BY-STEP SOLUTION

- With $u = 2x + 1$, we have $du = 2\,dx$, so $dx = du / 2$.
- Transforming the integration limits: when $x = 0$, $u = 2(0) + 1 = 1$; when $x = 4$, $u = 2(4) + 1 = 9$.
- Substituting into the integral: $\int_0^4 \sqrt{(2x+1)}\,dx = \int_1^9 \sqrt{u}\,(du/2) = (1/2)\int_1^9 \sqrt{u}\,du$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the differential factor 1/2 from $dx = du / 2$.

✗ **(B)** Transforms the integrand and differential but forgets to update the limits of integration.

✗ **(C)** Multiplies by 2 instead of dividing by 2.

Question 63 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) (e – 1) / 2

STEP-BY-STEP SOLUTION

- Let $u = x^2$, so $du = 2x\,dx$ and $x\,dx = du / 2$.
- For the bounds: when $x = 0$, $u = 0$; when $x = 1$, $u = 1$.
- The integral becomes: $(1/2)\int_0^1 e^u\,du = (1/2)[e^u]_0^1 = (1/2)(e^1 - e^0) = (1/2)(e - 1) = (e - 1)/2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the factor 1/2 from $du = 2x\,dx$.

✗ **(C)** Adds e^0 instead of subtracting: computes $(e + 1)/2$.

✗ **(D)** Assumes $e^0 = 0$ and computes $(e - 0)/2 = e/2$.

Question 64 Analysis

★★★★ MANDATORY

Correct Answer: (C) (1/5) $\sin^5 x + C$

STEP-BY-STEP SOLUTION

- Let $u = \sin x$, so $du = \cos x\,dx$.
- The integral becomes: $\int u^4\,du = u^5/5 + C = (1/5)\sin^5 x + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Substitutes $u = \cos x$ erroneously without adjusting for signs and powers.
✗ **(B)** Introduces an erroneous negative sign (the derivative of sine is positive cosine).
✗ **(D)** Differentiates using the Product Rule instead of integrating.

Question 65 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) (1/4) $(\ln x)^4 + C$

STEP-BY-STEP SOLUTION

- Let $u = \ln x$, so $du = (1/x)\,dx$.
- The integral transforms directly into: $\int u^3\,du = u^4/4 + C = (1/4)(\ln x)^4 + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to divide by the new power 4.
✗ **(C)** Does not increase the exponent.
✗ **(D)** Differentiates $(\ln x)^3/x$ using the Quotient Rule.

Question 66 Analysis

★★★★ MANDATORY

Correct Answer: (C) (1/4) $\tan^4 x + C$

STEP-BY-STEP SOLUTION

- Let $u = \tan x$, so $du = \sec^2 x\,dx$.
- The integral becomes: $\int u^3\,du = u^4/4 + C = (1/4)\tan^4 x + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Substitutes $u = \sec x$, which requires $\sec x \tan x\,dx$, not $\tan^3 x \sec^2 x\,dx$.
✗ **(B)** Fails to increment the exponent from 3 to 4.
✗ **(D)** Omits the denominator 4 from the power rule.

Question 67 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 1

STEP-BY-STEP SOLUTION

- Let $u = x^2 + 1$, so $du = 2x\,dx$ and $x\,dx = du / 2$.
- Limits: when $x = 0$, $u = 0^2 + 1 = 1$; when $x = \sqrt{3}$, $u = (\sqrt{3})^2 + 1 = 3 + 1 = 4$.
- The integral becomes: $(1/2) \int_1^4 u^{(-1/2)}\,du = (1/2)\,[2u^{(1/2)}]_1^4 = [\sqrt{u}]_1^4 = \sqrt{4} - \sqrt{1} = 2 - 1 = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets the factor 1/2 from $x\,dx = du/2$, giving $2(2 - 1) = 2$.
✗ **(C)** Multiplies by 1/2 without evaluating the upper bound square root properly.
✗ **(D)** Substitutes the original upper bound $\sqrt{3}$ into the antiderivative.

Question 68 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) (2/5)(x – 1)^(5/2) + (2/3)(x – 1)^(3/2) + C

STEP-BY-STEP SOLUTION

- Let $u = x - 1$, so $x = u + 1$ and $dx = du$.
- Substitute into the integral: $\int (u + 1)\sqrt{u}\,du = \int (u^{(3/2)} + u^{(1/2)})\,du = (2/5)u^{(5/2)} + (2/3)u^{(3/2)} + C$.
- Substituting back $u = x - 1$: $(2/5)(x - 1)^{(5/2)} + (2/3)(x - 1)^{(3/2)} + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Ignores the extra factor of x in the integrand.
✗ **(B)** Integrates x and $\sqrt{(x - 1)}$ separately and multiplies their antiderivatives.
✗ **(C)** Sign error: subtracts the second term instead of adding it.

Question 69 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) ln 4

STEP-BY-STEP SOLUTION

- Recognize that $f'(x)/f(x)$ is the derivative of $\ln(f(x))$.
- Let $u = f(x)$, $du = f'(x)\,dx$.
- When $x = 1$, $u = f(1) = 2$; when $x = 4$, $u = f(4) = 8$.
- The integral is: $\int_2^8 (1/u)\,du = [\ln|u|]_2^8 = \ln 8 - \ln 2 = \ln(8/2) = \ln 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Subtracts the arguments instead of dividing: $\ln(8 - 2) = \ln 6$.
✗ **(C)** Subtracts the outputs without applying the natural logarithm: $8/2 = 4$.
✗ **(D)** Computes $f(4) - f(1) = 8 - 2 = 6$.

Question 70 Analysis

★★★★ MANDATORY

Correct Answer: (C) 26/3

STEP-BY-STEP SOLUTION

- Let $u = g(x)$, so $du = g'(x)\,dx$.
- Transform the limits: when $x = 0$, $u = g(0) = 1$; when $x = 2$, $u = g(2) = 3$.
- The integral becomes: $\int_1^3 u^2\,du = [u^3/3]_1^3 = (3^3/3) - (1^3/3) = (27/3) - (1/3) = 26/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes $[g(2)]^2 - [g(0)]^2 = 9 - 1 = 8$.
✗ **(B)** Forgets to divide by 3 in the power rule: $27 - 1 = 26$.
✗ **(D)** Divides 26 by 6 instead of 3.

Question 71 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $x^2/2 - x + 2\ln|x+1| + C$

STEP-BY-STEP SOLUTION

- Since the degree of the numerator (2) is greater than the degree of the denominator (1), perform polynomial long division: $(x^2+1)/(x+1) = x - 1 + 2/(x+1)$.
- Integrate term by term: $\int (x - 1 + 2/(x+1))\,dx = x^2/2 - x + 2\ln|x+1| + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Performs division incorrectly as $x + 1$ with remainder 1.
✗ **(B)** Omits the degree-2 integration term $x^2/2$.
✗ **(D)** Integrates the numerator and denominator separately, which is mathematically invalid.

Question 72 Analysis

★★★★ MANDATORY

Correct Answer: (D) $3x - \ln|x+2| + C$

STEP-BY-STEP SOLUTION

- Rewrite the numerator by grouping: $3x + 5 = 3(x+2) - 1$.
- Thus, $(3x+5)/(x+2) = 3 - 1/(x+2)$.
- Integrating gives: $\int (3 - 1/(x+2))\,dx = 3x - \ln|x+2| + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error: adds $\ln|x+2|$ instead of subtracting.
✗ **(B)** Omits the linear polynomial term $3x$.
✗ **(C)** Uses the constant 5 as the coefficient of the logarithm instead of the true remainder -1.

Question 73 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) arctan(x + 2) + C

STEP-BY-STEP SOLUTION

- Complete the square in the denominator: $x^2 + 4x + 5 = (x^2 + 4x + 4) + 1 = (x + 2)^2 + 1$.
- The integral becomes: $\int (1 / ((x + 2)^2 + 1))\, dx$.
- Let $u = x + 2$, so $du = dx$.
- Then $\int (1 / (u^2 + 1))\, du = \arctan(u) + C = \arctan(x + 2) + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Mistakes the quadratic denominator for a logarithmic antiderivative (requires $2x + 4$ in the numerator).
✗ **(C)** Uses 5 instead of 1 in the inverse tangent argument and coefficient.
✗ **(D)** Does not divide the linear coefficient by 2 when completing the square.

Question 74 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) (1/2) arctan((x – 3)/2) + C

STEP-BY-STEP SOLUTION

- Complete the square: $x^2 - 6x + 13 = (x^2 - 6x + 9) + 4 = (x - 3)^2 + 2^2$.
- The integral matches the standard form $\int (1 / (u^2 + a^2))\, du = (1/a) \arctan(u/a) + C$ with $u = x - 3$ and $a = 2$.
- Thus, the antiderivative is $(1/2) \arctan((x - 3)/2) + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits the leading factor $1/a = 1/2$ outside the arctangent function.
✗ **(C)** Uses $a = 4$ instead of $a = \sqrt{4} = 2$.
✗ **(D)** Incorrectly uses a logarithmic formula.

Question 75 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) arcsin((x + 2)/3) + C

STEP-BY-STEP SOLUTION

- Complete the square inside the radical: $5 - 4x - x^2 = 5 - (x^2 + 4x) = 5 - [(x + 2)^2 - 4] = 9 - (x + 2)^2 = 3^2 - (x + 2)^2$.
- The integral becomes: $\int (1 / \sqrt{(3^2 - (x + 2)^2)})\, dx$.
- Using the standard arcsine formula $\int (1 / \sqrt{(a^2 - u^2)})\, du = \arcsin(u/a) + C$ with $u = x + 2$ and $a = 3$ gives $\arcsin((x + 2)/3) + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Includes an unnecessary constant factor 1/3 in front of arcsin.
✗ **(B)** Uses arctangent instead of arcsine.
✗ **(C)** Mistakes the radical for an inverse hyperbolic / log form.

Question 76 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 2 – ln 2

STEP-BY-STEP SOLUTION

- Perform polynomial long division: $(2x^2 + 3x) / (x + 1) = 2x + 1 - 1 / (x + 1)$.
- Integrate: $\int_0^1 (2x + 1 - 1/(x + 1))\, dx = [x^2 + x - \ln|x + 1|]_0^1$.
- Evaluating at $x = 1$: $1^2 + 1 - \ln(2) = 2 - \ln 2$.
- Evaluating at $x = 0$: $0 + 0 - \ln(1) = 0$.
- Subtracting gives $2 - \ln 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error on the logarithmic term: adds ln 2 instead of subtracting.
✗ **(C)** Evaluates $[x^2 + x]$ at $x = 1$ as 1 instead of $1 + 1 = 2$.
✗ **(D)** Doubles the logarithmic subtraction.

Question 77 Analysis

★★★★ MANDATORY

Correct Answer: (C) $\ln(x^2 + 1) + 5 \arctan x + C$

STEP-BY-STEP SOLUTION

- Split the integrand into two separate fractions: $\int (2x / (x^2 + 1))\, dx + \int (5 / (x^2 + 1))\, dx$.
- For the first term, let $u = x^2 + 1$, so $du = 2x\, dx$, giving $\int (1/u)\, du = \ln(x^2 + 1)$.
- For the second term, $5 \int (1 / (x^2 + 1))\, dx = 5 \arctan x$.
- Combining gives $\ln(x^2 + 1) + 5 \arctan x + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Keeps an extra factor of 2 in front of the natural log.
✗ **(B)** Divides 5 by 2 unnecessarily.
✗ **(D)** Includes a variable factor $2x$ in front of $\ln(x^2 + 1)$.

Question 78 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) arctan(2)

STEP-BY-STEP SOLUTION

- Complete the square: $x^2 + 2x + 2 = (x + 1)^2 + 1$.
- The integral is $\int_{-1}^{1} (1 / ((x + 1)^2 + 1))\, dx$.
- Let $u = x + 1$, so $du = dx$.
- Limits: when $x = -1$, $u = 0$; when $x = 1$, $u = 2$.
- The integral becomes: $\int_0^2 (1 / (u^2 + 1))\, du = [\arctan u]_0^2 = \arctan(2) - \arctan(0) = \arctan(2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses $\arctan(2)$ with $\arctan(1) = \pi/4$.
✗ **(B)** Evaluates the upper bound as $u = 1$ instead of $u = 2$.
✗ **(C)** Integrates using a natural log formula incorrectly.

Question 79 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $x^2/2 - (1/2)\ln(x^2+1) + 2\arctan x + C$

STEP-BY-STEP SOLUTION

- Perform polynomial division: $(x^3+2)/(x^2+1) = x + (-x+2)/(x^2+1) = x - x/(x^2+1) + 2/(x^2+1)$.
- Integrate term by term: $\int x\,dx = x^2/2$; $\int -(x/(x^2+1))\,dx = -(1/2)\ln(x^2+1)$; $\int (2/(x^2+1))\,dx = 2\arctan x$.
- Summing these terms yields: $x^2/2 - (1/2)\ln(x^2+1) + 2\arctan x + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets the 1/2 factor from substituting $u = x^2+1$ in the middle term.
✗ **(C)** Sign error on the middle logarithmic term.
✗ **(D)** Integrates x as x^2 instead of $x^2/2$.

Question 80 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\pi/6$

STEP-BY-STEP SOLUTION

- Complete the square inside the radical: $4x - x^2 = -(x^2 - 4x) = -[(x-2)^2 - 4] = 4 - (x-2)^2 = 2^2 - (x-2)^2$.
- The integral is $\int_1^2 (1/\sqrt{(2^2 - (x-2)^2)})\,dx = [\arcsin((x-2)/2)]_1^2$.
- Evaluating at the limits: At $x = 2$: $\arcsin((2-2)/2) = \arcsin(0) = 0$.
- At $x = 1$: $\arcsin((1-2)/2) = \arcsin(-1/2) = -\pi/6$.
- Subtracting gives: $0 - (-\pi/6) = \pi/6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates $\arcsin(1/2)$ as $\pi/3$ instead of $\pi/6$.
✗ **(C)** Confuses $\arcsin(1/2)$ with $\arcsin(\sqrt{2}/2) = \pi/4$.
✗ **(D)** Drops the negative sign when subtracting the lower limit: $0 - (\pi/6) = -\pi/6$.

Question 81 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $(1/2)\,x\,e^{2x} - (1/4)\,e^{2x} + C$

STEP-BY-STEP SOLUTION

- Use Integration by Parts: $\int u\,dv = uv - \int v\,du$.
- Choose $u = x$ and $dv = e^{2x}\,dx$.
- Then $du = dx$ and $v = (1/2)e^{2x}$.
- Applying the formula: $\int x\,e^{2x}\,dx = (1/2)x\,e^{2x} - \int (1/2)e^{2x}\,dx = (1/2)x\,e^{2x} - (1/4)e^{2x} + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error in the integration by parts formula: adds $\int v\,du$ instead of subtracting.
✗ **(C)** Omits both factors of 1/2 from integrating e^{2x}.
✗ **(D)** Integrates x and e^{2x} separately and multiplies their antiderivatives.

Question 82 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $(e^2+1)/4$

STEP-BY-STEP SOLUTION

- Apply Integration by Parts: choose $u = \ln x$ and $dv = x\,dx$.
- Then $du = (1/x)\,dx$ and $v = x^2/2$.
- The formula gives: $[(x^2/2)\ln x]_1^e - \int_1^e (x^2/2)(1/x)\,dx = [(x^2/2)\ln x]_1^e - (1/2)\int_1^e x\,dx = [(x^2/2)\ln x - x^2/4]_1^e$.
- Evaluate at $x = e$: $(e^2/2)(1) - e^2/4 = e^2/4$.
- Evaluate at $x = 1$: $(1/2)(0) - 1/4 = -1/4$.
- Subtracting gives: $e^2/4 - (-1/4) = (e^2+1)/4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error when evaluating the lower limit: subtracts 1/4 instead of adding.
✗ **(B)** Evaluates only the boundary term $[uv]_1^e = e^2/2$ and ignores the remaining integral.
✗ **(C)** Arithmetical slip in combining fractions.

Question 83 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $-x^2\cos x + 2x\sin x + 2\cos x + C$

STEP-BY-STEP SOLUTION

- Use Tabular Integration with signs (+, −, +): Differentiate $u = x^2$ repeatedly: $x^2 \to 2x \to 2 \to 0$.
- Integrate $dv = \sin x$ repeatedly: $\sin x \to -\cos x \to -\sin x \to \cos x$.
- Multiply along diagonals: $(+)(x^2)(-\cos x) + (-)(2x)(-\sin x) + (+)(2)(\cos x) + C = -x^2\cos x + 2x\sin x + 2\cos x + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error on the middle term: $(-)(2x)(-\sin x) = +2x\sin x$.
✗ **(C)** Reverses signs of all three terms.
✗ **(D)** Sign error on the constant cosine term: $(+)(2)(\cos x) = +2\cos x$.

Question 84 Analysis

★★★★ MANDATORY

Correct Answer: (C) $x\arctan x - (1/2)\ln(1+x^2) + C$

STEP-BY-STEP SOLUTION

- Set $u = \arctan x$ and $dv = dx$.
- Then $du = 1/(1+x^2)\,dx$ and $v = x$.
- Integration by Parts gives: $\int \arctan x\,dx = x\arctan x - \int (x/(1+x^2))\,dx$.
- For the remaining integral, use substitution with $w = 1+x^2$, $dw = 2x\,dx$: $\int (x/(1+x^2))\,dx = (1/2)\ln(1+x^2)$.
- Thus, the antiderivative is $x\arctan x - (1/2)\ln(1+x^2) + C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses the antiderivative with the derivative of arctan x.
✗ **(B)** Omits the coefficient 1/2 from $dw = 2x\,dx$.
✗ **(D)** Adds the integral term instead of subtracting.

Question 85 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $\ln|(x-3)/(x-2)|+C$

STEP-BY-STEP SOLUTION

- Factor the denominator: $x^2-5x+6=(x-2)(x-3)$.
- Decompose using partial fractions: $1/((x-2)(x-3))=A/(x-2)+B/(x-3)\Rightarrow 1=A(x-3)+B(x-2)$.
- At $x=2$: $1=-A\Rightarrow A=-1$.
- At $x=3$: $1=B\Rightarrow B=1$.
- Therefore: $\int(-1/(x-2)+1/(x-3))\,dx=-\ln|x-2|+\ln|x-3|+C=\ln|(x-3)/(x-2)|+C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Reverses the signs of the partial fraction constants A and B.
✗ **(C)** Adds both logarithmic terms instead of subtracting.
✗ **(D)** Attempts an invalid substitution treating the numerator as the derivative.

Question 86 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $(1/6)\ln(7/4)$

STEP-BY-STEP SOLUTION

- Factor: $x^2-9=(x-3)(x+3)$.
- Partial fractions: $1/((x-3)(x+3))=(1/6)[1/(x-3)-1/(x+3)]$.
- Antidifferentiate: $(1/6)\,[\ln|x-3|-\ln|x+3|\,]_4^5=(1/6)\,[\ln|(x-3)/(x+3)|\,]_4^5$.
- Evaluate at $x=5$: $(1/6)\ln(2/8)=(1/6)\ln(1/4)$.
- Evaluate at $x=4$: $(1/6)\ln(1/7)$.
- Subtracting gives: $(1/6)[\ln(1/4)-\ln(1/7)]=(1/6)\ln((1/4)/(1/7))=(1/6)\ln(7/4)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses partial fraction coefficients 1/3 instead of $1/(2\cdot3)=1/6$.
✗ **(C)** Inverts the logarithmic argument to 4/7.
✗ **(D)** Omits the leading partial fraction multiplier 1/6 entirely.

Question 87 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $\ln|x+1|-1/(x+1)+C$

STEP-BY-STEP SOLUTION

- Decompose by rewriting the numerator: $x+2=(x+1)+1$.
- Thus, $(x+2)/(x+1)^2=1/(x+1)+1/(x+1)^2$.
- Integrating term by term: $\int(1/(x+1)+(x+1)^{-2})\,dx=\ln|x+1|+((x+1)^{-1}/(-1))+C=\ln|x+1|-1/(x+1)+C$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error on the power rule integration of $(x+1)^{-2}$: adds $1/(x+1)$ instead of subtracting.
✗ **(B)** Uses coefficient 2 on the second term.
✗ **(C)** Treats the entire denominator as a single linear argument.

Question 88 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1/2

STEP-BY-STEP SOLUTION

- Express the improper integral as a limit: $\lim_{b\to\infty} \int_1^b x^{-3}\,dx = \lim_{b\to\infty} [-1/(2x^2)]_1^b = \lim_{b\to\infty} [(-1/(2b^2)) - (-1/(2(1)^2))] = 0 + 1/2 = 1/2$.
- Since the limit is a finite real number, the integral converges to 1/2.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the factor 2 in the denominator from integrating x^{-3}.

✗ **(B)** Inverts the result to 2.

✗ **(D)** Incorrectly concludes that integrals with infinite bounds always diverge.

Question 89 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 1/3

STEP-BY-STEP SOLUTION

- Rewrite as a limit: $\lim_{b\to\infty} \int_0^b e^{-3x}\,dx = \lim_{b\to\infty} [-(1/3)e^{-3x}]_0^b = \lim_{b\to\infty} [-(1/3)e^{-3b} - (-(1/3)e^0)]$.
- As $b\to\infty$, $e^{-3b}\to 0$.
- Therefore, the limit is $0 + (1/3)(1) = 1/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Multiplies by 3 instead of dividing by 3.

✗ **(C)** Evaluates only $e^0 = 1$ without the coefficient 1/3.

✗ **(D)** Claims the integral diverges due to the infinite upper limit.

Question 90 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 4

STEP-BY-STEP SOLUTION

- The integrand has an infinite discontinuity at the lower bound $x = 0$.
- Express as a limit: $\lim_{c\to 0^+} \int_c^4 x^{(-1/2)}\,dx = \lim_{c\to 0^+} [2\sqrt{x}]_c^4 = \lim_{c\to 0^+} (2\sqrt{4} - 2\sqrt{c}) = 2(2) - 0 = 4$.
- The integral converges to 4.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to multiply by 2 from the power rule: computes $\sqrt{4} - 0 = 2$.

✗ **(C)** Multiplies by 4 instead of 2.

✗ **(D)** Assumes the integral diverges because $1/\sqrt{x}$ has a vertical asymptote at $x = 0$.

Question 91: Area & Accumulation Function from a Graph Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

Part (a) Solution

- **Evaluating g(2):** By definition, $g(2) = \int_{-2}^{2} f(t)\,dt$, which represents the geometric area under the semicircle of radius r = 2. Area = $(1/2)\pi r^2 = (1/2)\pi(2^2) = 2\pi$.
- **Evaluating g(−4):** Reversing limits yields $g(-4) = \int_{-2}^{-4} f(t)\,dt = -\int_{-4}^{-2} f(t)\,dt$. Since f(t) = 0 along the t-axis for all t in [−4, −2], the integral is 0.
- **Final Result:** $g(2) = 2\pi$ and $g(-4) = 0$.

Part (b) Solution

- **Finding g'(4):** By the Fundamental Theorem of Calculus Part 1, g'(x) = f(x). Therefore, g'(4) = f(4) = 4.
- **Derivative of f at x = 4:** g"(4) = f'(4). The graph of f consists of line segments meeting at a corner at (4, 4).
- **One-Sided Derivatives:** The left-hand derivative is $\lim_{x\to 4^-} f'(x) = (4-0)/(4-2) = 2$, while the right-hand derivative is $\lim_{x\to 4^+} f'(x) = (0-4)/(6-4) = -2$. Because these limits differ, g"(4) does not exist.

Part (c) Solution

- **Critical Point Condition:** Relative extrema can only occur where g'(x) = f(x) = 0 in the interior (−4, 6). The roots of f are x = −2 and x = 2.
- **Testing x = −2:** For x < −2, f(x) = 0; for x > −2, f(x) > 0. The derivative does not transition from negative to positive.
- **Testing x = 2:** For x < 2, f(x) > 0 (semicircle); for x > 2, f(x) > 0 (triangle). Because g' does not change sign at x = 2, there is no relative minimum anywhere in (−4, 6).

Part (d) Solution

- **Monotonicity of g:** Since the graph of f lies entirely on or above the t-axis for all t in [−4, 6], we have g'(x) = f(x) ≥ 0 everywhere. This guarantees that g is nondecreasing on [−4, 6].
- **Evaluating Endpoints:** Left endpoint: g(−4) = 0. Right endpoint: $g(6) = \int_{-2}^{6} f(t)\,dt$ = Area(semicircle) + Area(triangle on [2, 6]).
- **Calculation of Maximum Value:** $g(6) = 2\pi + (1/2)(\text{base})(\text{height}) = 2\pi + (1/2)(4)(4) = 2\pi + 8$. The absolute maximum value of g on [−4, 6] is $2\pi + 8$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Computes g(2) = 2π and g(−4) = 0 with geometric setups	**2 pts**
Part (b): Identifies g'(4) = 4 (1 pt) and explains why g"(4) does not exist (1 pt)	**2 pts**
Part (c): Considers f(x) = 0 and justifies that no relative minimum exists in (−4, 6)	**2 pts**
Part (d): Considers endpoint x = 6, computes area = 2π + 8, and justifies absolute maximum	**3 pts**

Question 92: Tabular Rates & Riemann Sum Approximations Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

Part (a) Solution

- **Formula Setup:** For non-uniform subintervals, compute each trapezoid individually: $T = \sum \Delta t_i \cdot [\,(R(t_{i-1}) + R(t_i))\,/\,2\,]$.
- **Subinterval Computations:**
  - $[0, 3]$: $(3 - 0) \cdot [(4 + 7)/2] = 3(5.5) = 16.5$
  - $[3, 5]$: $(5 - 3) \cdot [(7 + 9)/2] = 2(8.0) = 16.0$
  - $[5, 8]$: $(8 - 5) \cdot [(9 + 12)/2] = 3(10.5) = 31.5$
  - $[8, 12]$: $(12 - 8) \cdot [(12 + 15)/2] = 4(13.5) = 54.0$
- **Summation:** Total Volume $\approx 16.5 + 16.0 + 31.5 + 54.0 = 118$ gallons.

Part (b) Solution

- **Integral Interpretation:** The expression $(1 / (12 - 0)) \int_0^{12} R(t)\,dt$ represents the average value of the rate function $R(t)$ over the interval $[0, 12]$.
- **Contextual Meaning:** It is the average rate at which water is pumped into the storage tank over the 12-hour period from $t = 0$ to $t = 12$ hours.
- **Units:** The units are gallons per hour (gal/hr).

Part (c) Solution

- **Interval Selection:** The value $t = 6$ lies between the given table values $t = 5$ and $t = 8$.
- **Difference Quotient:** $R'(6) \approx (R(8) - R(5)) / (8 - 5) = (12 - 9) / 3 = 3 / 3 = 1$.
- **Units & Conclusion:** $R'(6) \approx 1$ gallon per hour squared (gal/hr^2).

Part (d) Solution

- **Over/Underestimate Justification:** Because $R(t)$ is strictly increasing, the function achieves its maximum value at the right endpoint of every subinterval. Therefore, a Right Riemann sum is an overestimate.
- **Right Riemann Calculation:** $R = 3(7) + 2(9) + 3(12) + 4(15) = 21 + 18 + 36 + 60 = 135$ gallons.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct trapezoidal setups with subinterval widths (2 pts) and answer 118 gallons (1 pt)	**3 pts**
Part (b): Meaning as average pumping rate (1 pt) with correct units gal/hr (1 pt)	**2 pts**
Part (c): Difference quotient setup and value 1 (1 pt) with units gal/hr^2 (1 pt)	**2 pts**
Part (d): States overestimate with justification (1 pt) and evaluates Right Riemann sum = 135 (1 pt)	**2 pts**

Question 93: Fundamental Theorem of Calculus with Composite Bounds Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

Part (a) Solution

- **Upper Limit Evaluation:** Substituting $x = 1/2$ into the upper limit gives $2(1/2) = 1$.
- **Property of Integrals:** $H(1/2) = \int_{1}^{1} \sqrt{(t^2 + 5)}\, dt$. Whenever the upper and lower limits of integration are identical, the value of the definite integral is zero.
- **Final Result:** $H(1/2) = 0$.

Part (b) Solution

- **Chain Rule with FTC 1:** By the Leibniz Rule, $\frac{d}{dx}\left[\int_{1}^{u(x)} f(t)\, dt\right] = f(u(x)) \cdot u'(x)$.
- **Derivative Expression:** Here $u(x) = 2x$, so $u'(x) = 2$. Thus: $H'(x) = \sqrt{((2x)^2 + 5)} \cdot 2 = 2\sqrt{(4x^2 + 5)}$.
- **Evaluating at x = 2:** $H'(2) = 2\sqrt{(4(2^2) + 5)} = 2\sqrt{(16 + 5)} = 2\sqrt{21}$.

Part (c) Solution

- **Point of Tangency:** At $x = 1/2$, from part (a), $y = H(1/2) = 0$.
- **Numerical Slope:** $H'(1/2) = 2\sqrt{(4(1/4) + 5)} = 2\sqrt{(1 + 5)} = 2\sqrt{6}$.
- **Tangent Line Equation:** $y - 0 = 2\sqrt{6}\,(x - 1/2) \Rightarrow y = 2\sqrt{6}\,(x - 1/2)$.

Part (d) Solution

- **Computing Second Derivative:** Differentiate $H'(x) = 2(4x^2 + 5)^{(1/2)}$ using the Chain Rule: $H''(x) = 2 \cdot (1/2)(4x^2 + 5)^{(-1/2)} \cdot (8x) = 8x / \sqrt{(4x^2 + 5)}$.
- **Evaluating at x = 2:** $H''(2) = 8(2) / \sqrt{(4(4) + 5)} = 16 / \sqrt{21}$.
- **Concavity Conclusion:** Since $H''(2) = 16/\sqrt{21} > 0$, the graph of H is concave up at $x = 2$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Evaluates $H(1/2) = 0$ with justification	**2 pts**
Part (b): Applies Chain Rule to find $H'(x)$ (2 pts) and evaluates $H'(2) = 2\sqrt{21}$ (1 pt)	**3 pts**
Part (c): Computes slope $H'(1/2) = 2\sqrt{6}$ and writes tangent line equation	**2 pts**
Part (d): Computes $H''(x)$ correctly (1 pt) and concludes concave up with justification (1 pt)	**2 pts**

Question 94: Definite Integration with Substitution & Particle Motion Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

Part (a) Solution

- **Derivative of Velocity:** Acceleration is $a(t) = v'(t)$. Differentiate $v(t) = t(t^2 + 9)^{(-1/2)}$ using the Product or Quotient Rule.
- **Simplification:** $a(t) = [\,(1)\cdot\sqrt{(t^2 + 9)} - t \cdot (t / \sqrt{(t^2 + 9)})\,] / (t^2 + 9) = [\,(t^2 + 9) - t^2\,] / (t^2 + 9)^{(3/2)} = 9 / (t^2 + 9)^{(3/2)}$.

Part (b) Solution

- **Substitution Setup:** Let $u = t^2 + 9$, so $du = 2t\,dt$, which means $t\,dt = du / 2$.
- **Bound Transformation:** Lower bound: when $t = 0$, $u = 0^2 + 9 = 9$. Upper bound: when $t = 4$, $u = 4^2 + 9 = 25$.
- **Evaluation:** $(1/2)\ \int_{9}^{25} u^{(-1/2)}\,du = (1/2)\,[\,2u^{(1/2)}\,]_9^{25} = [\,\sqrt{u}\,]_9^{25} = \sqrt{25} - \sqrt{9} = 5 - 3 = 2$ meters.

Part (c) Solution

- **Net Change Theorem:** $s(4) = s(0) + \int_0^4 v(t)\,dt$.
- **Position Calculation:** Given $s(0) = 5$ meters and displacement = 2 meters: $s(4) = 5 + 2 = 7$ meters.

Part (d) Solution

- **Average Value Formula:** $v_{avg} = (1 / (4 - 0))\ \int_0^4 v(t)\,dt$.
- **Calculation:** $v_{avg} = (1/4)(2) = 1/2$ meter per second (m/s).

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Differentiates v(t) using Quotient or Product Rule to obtain $a(t) = 9/(t^2+9)^{(3/2)}$	**2 pts**
Part (b): Identifies $u = t^2+9$ and du (1 pt), transforms limits to [9, 25] (1 pt), answer 2 (1 pt)	**3 pts**
Part (c): Adds initial position to displacement: $s(4) = 5 + 2 = 7$	**2 pts**
Part (d): Sets up average value integral and evaluates to 1/2 m/s	**2 pts**

Question 95: Integration by Parts & Improper Integrals (BC Special) Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

Part (a) Solution

- **Tabular Setup:** Use alternating signs (+, −, +, −) with derivatives of $u = x^2$ and antiderivatives of $dv = e^{-x}\,dx$.
- **Derivatives of u:** $x^2 \to 2x \to 2 \to 0$.
- **Antiderivatives of dv:** $e^{-x} \to -e^{-x} \to e^{-x} \to -e^{-x}$.
- **Diagonal Pairing:** $(+)(x^2)(-e^{-x}) + (-)(2x)(e^{-x}) + (+)(2)(-e^{-x}) + C = -e^{-x}(x^2 + 2x + 2) + C$.

Part (b) Solution

- **Antiderivative Function:** Let $F(x) = -e^{-x}(x^2 + 2x + 2)$.
- **Boundary Evaluations:** At $x = 2$: $F(2) = -e^{-2}(2^2 + 2(2) + 2) = -10e^{-2}$. At $x = 0$: $F(0) = -e^{0}(0 + 0 + 2) = -2$.
- **Definite Value:** $F(2) - F(0) = -10e^{-2} - (-2) = 2 - 10e^{-2} = 2 - 10/e^2$.

Part (c) Solution

- **Improper Limit Setup:** $\int_0^\infty x^2 e^{-x}\,dx = \lim_{b\to\infty} \int_0^b x^2 e^{-x}\,dx = \lim_{b\to\infty} [-e^{-b}(b^2 + 2b + 2) - (-2)]$.
- **Limit Evaluation:** By dominance of exponential decay (or repeated L'Hôpital's Rule), $\lim_{b\to\infty} (b^2 + 2b + 2)/e^b = 0$.
- **Conclusion:** The limit evaluates to $0 + 2 = 2$. The improper integral converges to 2.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Sets up Integration by Parts (2 pts) and completes to $-e^{-x}(x^2+2x+2) + C$ (2 pts)	**4 pts**
Part (b): Evaluates limits correctly to obtain $2 - 10/e^2$	**3 pts**
Part (c): Writes limit setup (1 pt) and evaluates limit to 2 using L'Hôpital / dominance (1 pt)	**2 pts**

Question 96: Partial Fractions & Initial Value Differential Problem (BC Special) Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

Part (a) Solution

- **Factoring Denominator:** $x^2 - x - 6 = (x - 3)(x + 2)$.
- **Partial Fractions Template:** $(x - 5) / ((x - 3)(x + 2)) = A / (x - 3) + B / (x + 2) \Rightarrow x - 5 = A(x + 2) + B(x - 3)$.
- **Solving Constants:** Setting $x = 3$ gives $3 - 5 = 5A \Rightarrow A = -2/5$. Setting $x = -2$ gives $-2 - 5 = -5B \Rightarrow B = 7/5$.
- **Result:** $(x - 5)/(x^2 - x - 6) = (-2/5)/(x - 3) + (7/5)/(x + 2)$.

Part (b) Solution

- **Integration Step:** $y = \int [(-2/5)/(x - 3) + (7/5)/(x + 2)]\, dx = -(2/5) \ln|x - 3| + (7/5) \ln|x + 2| + C$.
- **Domain Note:** For $x > 3$, both $(x - 3) > 0$ and $(x + 2) > 0$, so $y = -(2/5) \ln(x - 3) + (7/5) \ln(x + 2) + C$.
- **Applying Initial Condition y(4) = ln 2:** $\ln 2 = -(2/5) \ln(1) + (7/5) \ln(6) + C \Rightarrow C = \ln 2 - (7/5) \ln 6$.
- **Explicit Solution:** $y(x) = -(2/5) \ln(x - 3) + (7/5) \ln(x + 2) + \ln 2 - (7/5) \ln 6$.

Part (c) Solution

- **Combining Logarithmic Terms:** $y(x) = \ln [(x + 2)^{(7/5)} / (x - 3)^{(2/5)}] + C$.
- **Limit Analysis:** As $x \to \infty$, the ratio behaves asymptotically as $x^{(7/5)} / x^{(2/5)} = x^{(5/5)} = x \to \infty$.
- **Conclusion:** $\lim_{x \to \infty} y(x) = \infty$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Sets up partial fractions (1 pt) and solves A = -2/5, B = 7/5 (2 pts)	**3 pts**
Part (b): Integrates to logarithmic terms (2 pts), applies initial condition y(4) = ln 2 (1 pt), explicit solution (1 pt)	**4 pts**
Part (c): Evaluates limit of combined logarithmic expression to ∞	**2 pts**

Question 97: Improper Integrals Analysis & Parameter Investigation (BC Special) Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

Part (a) Solution

- **Limit Definition:** $I(2) = \lim_{b\to\infty} \int_1^b x^{-2}\,dx$.
- **Antiderivative:** $\lim_{b\to\infty} [-1/x]_1^b = \lim_{b\to\infty} [(-1/b) - (-1/1)] = \lim_{b\to\infty} (1 - 1/b)$.
- **Convergence:** As $b \to \infty$, $1/b \to 0$. Therefore, the limit is $1 - 0 = 1$. The integral converges to 1.

Part (b) Solution

- **Limit Definition:** $I(1) = \lim_{b\to\infty} \int_1^b (1/x)\,dx$.
- **Antiderivative:** $\lim_{b\to\infty} [\ln x]_1^b = \lim_{b\to\infty} (\ln b - \ln 1) = \lim_{b\to\infty} \ln b$.
- **Divergence:** Because $\ln b \to \infty$ as $b \to \infty$, the limit does not exist. The integral diverges.

Part (c) Solution

- **Antiderivative for $p \neq 1$:** $\lim_{b\to\infty} \int_1^b x^{(-p)}\,dx = \lim_{b\to\infty} [x^{(1-p)}/(1-p)]_1^b = \lim_{b\to\infty} [(b^{(1-p)} - 1)/(1-p)]$.
- **Case 1 ($p > 1$):** Here $1 - p < 0$, so $b^{(1-p)} = 1/b^{(p-1)} \to 0$ as $b \to \infty$. The limit evaluates to $(0 - 1)/(1 - p) = 1/(p - 1)$, which is a finite real number (converges).
- **Case 2 ($0 < p < 1$):** Here $1 - p > 0$, so $b^{(1-p)} \to \infty$ as $b \to \infty$ (diverges). Combined with $p = 1$, $I(p)$ converges if and only if $p > 1$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Writes limit setup (1 pt), finds antiderivative (1 pt), concludes converges to 1 (1 pt)	**3 pts**
Part (b): Writes limit setup (1 pt), integrates to $\ln b$ (1 pt), concludes diverges to ∞ (1 pt)	**3 pts**
Part (c): Evaluates general antiderivative for $p \neq 1$ (1 pt), analyzes sign of $1 - p$ (1 pt), concludes converges for $p > 1$ (1 pt)	**3 pts**

Question 98: Definite Integrals: Symmetry, Properties & Piecewise Analysis Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

Part (a) Solution

- **Even Function Definition:** By symmetry across the y-axis, $f(-x) = f(x)$.
- **Calculation:** $\int_{-6}^{6} f(x)\,dx = 2\int_{0}^{6} f(x)\,dx = 2(10) = 20$.

Part (b) Solution

- **Linearity Property:** $\int_{-6}^{6} (3f(x) + 4)\,dx = 3\int_{-6}^{6} f(x)\,dx + \int_{-6}^{6} 4\,dx$.
- **Odd Function Symmetry:** Because f is odd, $\int_{-6}^{6} f(x)\,dx = 0$.
- **Constant Integration:** $\int_{-6}^{6} 4\,dx = 4 \cdot (6 - (-6)) = 4(12) = 48$. Total: $3(0) + 48 = 48$.

Part (c) Solution

- **Substitution Setup:** Let $u = 2x$, so $du = 2\,dx$ and $dx = du / 2$.
- **Transformed Limits:** Lower bound: when $x = 0$, $u = 0$. Upper bound: when $x = 3$, $u = 2(3) = 6$.
- **Evaluation:** $\int_{0}^{3} f(2x)\,dx = \int_{0}^{6} f(u)\,(du / 2) = (1/2)\int_{0}^{6} f(u)\,du = (1/2)(10) = 5$.

Part (d) Solution

- **Interval Additivity:** $\int_{0}^{6} f(x)\,dx = \int_{0}^{2} f(x)\,dx + \int_{2}^{6} f(x)\,dx$.
- **Solving for Unknown Integral:** $10 = \int_{0}^{2} f(x)\,dx + 7 \Rightarrow \int_{0}^{2} f(x)\,dx = 10 - 7 = 3$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Applies even symmetry property and evaluates to 20	**2 pts**
Part (b): Identifies odd integral = 0 (1 pt) and evaluates constant integral to 48 (1 pt)	**2 pts**
Part (c): Substitution $u = 2x$ and $du = 2\,dx$ with transformed bounds (2 pts) and answer 5 (1 pt)	**3 pts**
Part (d): Applies interval additivity to find $\int_{0}^{2} f(x)\,dx = 3$	**2 pts**

Question 99: Accumulation Function with Area & Analytic Pieces Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

Part (a) Solution

- **Evaluating g(4):** $g(4) = \int_0^4 \sqrt{(16 - t^2)}\, dt$ represents the area of a quarter circle of radius r = 4. Area = $(1/4)\pi(4^2) = 4\pi$.
- **Evaluating g(8):** $g(8) = g(4) + \int_4^8 (4 - t)\, dt$. On [4, 8], f(t) = 4 – t is a line segment forming a right triangle below the t-axis from (4, 0) to (8, –4): Area = –(1/2)(base)(height) = –(1/2)(4)(4) = –8.
- **Result:** $g(4) = 4\pi$ and $g(8) = 4\pi - 8$.

Part (b) Solution

- **Critical Point Definition:** Critical points occur where g'(x) = f(x) = 0 or does not exist.
- **Finding Zeros:** On (0, 4), $f(x) = \sqrt{(16 - x^2)} > 0$. At x = 4, f(4) = 0. On (4, 8), f(x) = 4 – x < 0.
- **Conclusion:** The only critical point in the open interval (0, 8) is x = 4.

Part (c) Solution

- **Candidate Evaluation:** Test boundaries and critical points on [0, 8]:
  - x = 0: g(0) = 0
  - x = 4: $g(4) = 4\pi \approx 12.566$
  - x = 8: $g(8) = 4\pi - 8 \approx 4.566$
- **Sign of Derivative:** g'(x) > 0 for 0 < x < 4 (g is strictly increasing) and g'(x) < 0 for 4 < x < 8 (g is strictly decreasing).
- **Conclusion:** The absolute maximum value of g on [0, 8] is 4π, occurring at x = 4.

Part (d) Solution

- **Slope at x = 4:** By FTC 1, g'(4) = f(4) = 0.
- **Point on Graph:** (4, g(4)) = (4, 4π).
- **Horizontal Tangent Equation:** $y - 4\pi = 0(x - 4) \Rightarrow y = 4\pi$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Computes g(4) = 4π (1 pt) and g(8) = 4π - 8 (1 pt)	**2 pts**
Part (b): Solves g'(x) = f(x) = 0 to find critical point x = 4	**2 pts**
Part (c): Evaluates candidates and justifies absolute maximum = 4π at x = 4	**3 pts**
Part (d): Identifies slope g'(4) = 0 and writes horizontal tangent line y = 4π	**2 pts**

Question 100: Comprehensive Mixed Integration Challenge (BC Culmination) Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

Part (a) Solution

- **Completing the Square:** $x^2 + 4x + 8 = (x^2 + 4x + 4) + 4 = (x + 2)^2 + 2^2$.
- **Standard Form:** The integral matches $\int (1 / (u^2 + a^2))\, du = (1/a) \arctan(u/a) + C$ with $u = x + 2$ and $a = 2$.
- **Antiderivative:** $\int (1 / ((x + 2)^2 + 4))\, dx = (1/2) \arctan((x + 2)/2) + C$.

Part (b) Solution

- **Limit Definition:** $\int_{-2}^{\infty} f(x)\, dx = \lim_{b \to \infty} \int_{-2}^{b} (1 / ((x + 2)^2 + 4))\, dx$.
- **Antiderivative Evaluation:** $\lim_{b \to \infty} [(1/2) \arctan((x + 2)/2)]_{-2}^{b} = \lim_{b \to \infty} [(1/2) \arctan((b + 2)/2) - (1/2) \arctan(0)]$.
- **Limit Value:** As $b \to \infty$, $(b + 2)/2 \to \infty$, so $\arctan((b + 2)/2) \to \pi/2$. Since $\arctan(0) = 0$, the limit is $(1/2)(\pi/2) - 0 = \pi/4$.

Part (c) Solution

- **Substitution Choice:** Let $u = x^2 + 4x + 8$, then $du = (2x + 4)\, dx$.
- **Limits of Integration:** Lower bound: $x = -2 \Rightarrow u = (-2)^2 + 4(-2) + 8 = 4$. Upper bound: $x = 0 \Rightarrow u = 0^2 + 4(0) + 8 = 8$.
- **Evaluation:** $\int_{4}^{8} (1/u)\, du = [\ln u]_{4}^{8} = \ln 8 - \ln 4 = \ln(8/4) = \ln 2$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Completes the square to $(x+2)^2+4$ (1 pt) and obtains $(1/2) \arctan((x+2)/2) + C$ (2 pts)	**3 pts**
Part (b): Sets up limit as $b \to \infty$ (1 pt), evaluates arctan limits (1 pt), answer $\pi/4$ (1 pt)	**3 pts**
Part (c): Substitution $u = x^2+4x+8$ and $du = (2x+4)dx$ (1 pt), limits [4, 8] (1 pt), answer $\ln 2$ (1 pt)	**3 pts**

Essential Unit 6 Rules

Core formulas required for mastery of accumulation, antiderivatives, and integration procedures.

- **FTC Part 1:** $d/dx\left[\int_a^x f(t)\,dt\right] = f(x)$. With chain rule: $d/dx\left[\int_a^{u(x)} f(t)\,dt\right] = f(u(x)) \cdot u'(x)$.
- **FTC Part 2 (Net Change):** $\int_a^b f'(x)\,dx = f(b) - f(a)$, or $f(b) = f(a) + \int_a^b f'(x)\,dx$.
- **Integration by Parts (BC):** $\int u\,dv = uv - \int v\,du$. Tabular method applies to polynomial × exponential/sine/cosine.
- **Partial Fractions (BC):** $1/((x-a)(x-b)) = A/(x-a) + B/(x-b)$. Antiderivative integrates to logarithms.
- **Improper Integrals (BC):** $\int_a^\infty f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx$. The p-integral converges if and only if $p > 1$.
- **Riemann Sum Limit:** $\lim_{n\to\infty} \sum_{k=1}^{n} f(a + k\Delta x)\,\Delta x = \int_a^b f(x)\,dx$, where $\Delta x = (b-a)/n$.
- **Trapezoidal Rule:** $T = (\Delta x/2)\left[f(x_0) + 2f(x_1) + 2f(x_2) + \ldots + 2f(x_{n-1}) + f(x_n)\right]$.
- **Error Bounds by Concavity:** If $f''(x) > 0$ (concave up), T > Integral (overestimate). If $f''(x) < 0$ (concave down), T < Integral (underestimate).
- **Inverse Trig Forms:** $\int (1/\sqrt{a^2 - u^2})\,du = \arcsin(u/a) + C$; $\int (1/(a^2 + u^2))\,du = (1/a)\arctan(u/a) + C$.

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Thank You, Students!

Mastery Comes Through Deliberate Practice

Congratulations on completing this 100-Question Priority-Starred Master Repository for Unit 6! Integration is the foundation of calculus. Review your errors, internalize the distractor rationales, and approach your AP Calculus BC examination with absolute confidence.

TARGET SCORE: 5 · AP CALCULUS BC